

PWM-LDCT v0.5: A Content-Addressed, Multi-Task Harmonization of the LIDC-IDRI Low-Dose CT Dataset for Reproducible Reconstruction Benchmarking

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Abstract

LIDC-IDRI, the most widely used public thoracic CT collection, anchors a large fraction of low-dose CT reconstruction research, yet it is distributed in a format-specific layout with heterogeneous annotation conventions and no content-addressed versioning. We present PWM-LDCT v0.5, a unified, content-addressed harmonization of the LIDC-IDRI collection as a first release. As new openly-licensed data records it deposits: harmonized multi-task annotations (majority-vote lung-nodule labels reused from the four-radiologist LIDC annotations, with the raw per-reader labels preserved); physically-grounded simulated low-dose images at dose ratios 0.10 / 0.25 / 0.50 for the full-dose-only scans, with the simulation discrepancy quantified in Technical Validation; harmonized per-scan metadata; and deterministic patient-level splits. These are delivered through a pip-installable Python loader and three Docker pipelines that deterministically regenerate the HDF5 shards from the source’s raw distribution (verified against a released per-voxel tolerance), and demonstrated by baseline reproductions of four reconstruction methods. The release is identified by a SHA-256 manifest hash and deposited under a PhysioNet DOI with a mirrored Zenodo archive. The AAPM 2016 low-dose collection, which adds genuine paired-dose coverage, is planned as a 10-patient real-paired validation subset of the v1.0 roadmap and is not part of this v0.5 release.

Background & Summary

Low-dose CT reconstruction aims to reduce the radiation exposure associated with CT imaging (Brenner and Hall, 2007), and public datasets have driven a decade of progress in deep-learning approaches to it (Chen et al., 2017; Wang et al., 2020; Diwakar and Kumar, 2018; Willemink and Noël, 2019). LIDC-IDRI (Armato III et al., 2011) (1,018 thoracic CT scans with four-radiologist nodule annotations; full-dose only) is the most widely used public thoracic CT collection and anchors a large fraction of the published evaluation literature. One further public low-dose CT collection — the 2016 AAPM-Mayo Low-Dose CT Grand Challenge release (AAPM, 2016) (10 patients with real full-dose acquisitions and matched 25%-dose data generated by Mayo’s validated projection-domain noise-insertion technique, not a second physical scan) — is planned for the PWM-LDCT v1.0 roadmap as a 10-patient real-paired validation subset rather than this first release, because it carries paired-dose coverage that requires additional licensing and harmonization work. The Mayo LDCT-and-Projection-Data collection (McCollough et al., 2017) (299 patients extending

AAPM 2016 with raw projection data; 199 publicly accessible, spanning GE and Siemens scanners) is retained as landscape context only and is not in the release scope. **Scope decision (2026-08-21): the v1.0 roadmap now adds only the AAPM 2016 collection (10 training patients, FD+QD paired) as a real-paired validation subset; the Mayo LDCT-PD collection and prospective multi-vendor acquisition are not in the release scope.** This v0.5 paper therefore focuses on the LIDC-IDRI collection alone.

Yet even this single, widely-used source is not designed to be used as a benchmark. Three structural gaps prevent the field from treating it as a coherent one:

1. **Loader fragmentation.** LIDC-IDRI is distributed in its own format. Every published method that evaluates on it re-implements its own preprocessing, leading to silent format-conversion drift between papers. A method reporting PSNR on “LIDC” may have used any of half a dozen subtly-different pixel-spacing, HU-windowing, or slice-thickness conventions.
2. **Annotation heterogeneity.** LIDC-IDRI ships four-radiologist nodule annotations in an XML format unique to the project. A method that wants to evaluate downstream-task performance on the full-dose images must reconcile this annotation convention with its own tooling; there is no shared, versioned majority-vote consolidation.
3. **Benchmark drift.** LIDC-IDRI has no content-addressed version identifier. As papers are written, the dataset is silently re-cleaned, re-split, or sub-sampled; a result tied to “LIDC-IDRI” in one paper is not guaranteed to be evaluated against the same scans as a result tied to “LIDC-IDRI” in another paper a year later.

PWM-LDCT v0.5 closes these three gaps by adding an annotation-and-infrastructure layer over the existing LIDC-IDRI collection rather than collecting new scans. Its *new, openly-licensed data records* are (a) harmonized multi-task annotations — majority-vote lung-nodule labels reused from the existing four-radiologist LIDC annotations, plus the raw per-reader labels behind each majority vote; (b) physically-grounded simulated low-dose images at dose ratios $r \in \{0.10, 0.25, 0.50\}$ for the full-dose-only LIDC scans, whose simulation behavior is quantified in Technical Validation rather than asserted; and (c) harmonized per-scan metadata with deterministic patient-level splits. These records are delivered through (d) a unified Python loader that surfaces the LIDC collection under a single schema and (e) Docker preprocessing pipelines that deterministically regenerate the harmonized HDF5 shards from the source’s raw distribution (reproducible within a released per-voxel tolerance), and they are version-pinned by (f) a content-addressed manifest identified by a SHA-256 hash that any downstream user can verify offline against a locally-built copy and that is deposited under a permanent PhysioNet DOI with a mirrored Zenodo archive. Baseline reproductions of four community-standard methods, reported in Technical Validation, demonstrate that the records support end-to-end training and evaluation without code modification. Table 1 summarizes the source composition of v0.5 and the v1.0 roadmap.

To situate the release against the wider landscape of public low-dose CT resources, Table 2 compares PWM-LDCT v0.5 with LIDC-IDRI (its source), the AAPM 2016 collection (planned as the v1.0 real-paired validation subset), the Mayo LDCT-PD collection (retained as landscape context only; not in the release scope), and the National Lung Screening Trial (NLST) (National Lung Screening Trial Research Team, 2011; National Cancer Institute, 2013), the largest publicly distributed low-dose CT screening collection. The comparison highlights what no prior public resource offers together: a *content-addressed, harmonized, multi-task* LIDC-derived benchmark with deterministic patient-level splits, plus a v1.0 plan that pairs *cross-vendor* simulated coverage with an AAPM 2016 (Siemens) real-paired validation subset under the same schema. NLST, while orders

Source	Patients	Paired LD ref.	Sinograms	Annotations as-distributed
LIDC-IDRI (Armato III et al., 2011) (v0.5)	1,010	No	No	4-radiologist nodule bbox/seg (XML)
<i>v1.0 validation subset (planned; not in v0.5)</i>				
AAPM 2016 (AAPM, 2016)	10	Yes	Yes	None as-distributed

Table 1: Composition of the PWM-LDCT v0.5 release. v0.5 is the LIDC-IDRI-only first release (1,010 patients available in the public distribution used here; the canonical LIDC-IDRI collection nominally has 1,018 patients, 8 missing from this distribution and noted in limitations). AAPM 2016 (10 patients) is planned as the v1.0 real-paired validation subset (see manuscript_v1.tex) and is **not** part of v0.5; its paired-dose coverage will be added in that release. Mayo LDCT-PD and prospective multi-vendor acquisition are **not** in the v1.0 scope (removed 2026-08-21).

of magnitude larger in participant count, provides single-acquisition screening scans without paired dose, without pixel-level lesion annotations, and without raw projection data; each of the listed collections provides only a subset of the needed dimensions (Table 2).

The end-to-end structure of the release — from the LIDC-IDRI source through the preprocessing pipeline and unified loader to the harmonized, content-addressed records — is summarized in Figure 1.

The v0.5 paper’s contribution is the new harmonized annotation and simulated-low-dose records, the unified loader and preprocessing layer, and the content-addressed manifest deposited under a PhysioNet DOI — not the underlying scans. The scans themselves already exist in the cited source, and v0.5 does not redistribute them under a different license; downstream users still obtain access to each source through that source’s own gatekeeper (NBIA for LIDC). What v0.5 *deposits* is the value-added layer: the new annotation records, the simulated-low-dose images derived from the public full-dose scans, the harmonized metadata, and the code that deterministically reproduces the harmonized shards from the source within a released per-voxel tolerance.

All descriptor claims (the harmonization layer, content-addressing, annotation reuse, the loader, and baseline reproducibility) are self-contained; separate companion work on evaluation methodology ([to be confirmed, 2027a,b](#)) and a community leaderboard ([to be confirmed, 2028](#)) is in preparation but is not required to obtain, use, or validate this release.

Methods

Ethics, consent, and data sharing

PWM-LDCT v0.5 does not collect new data. All scans are sourced from the LIDC-IDRI public dataset, which was released under its institutional approval. Our use complies with the data-use agreement of the source:

- **LIDC-IDRI** ([Armato III et al., 2011](#)): CC BY 3.0 with attribution requirement; no redistribution restriction. PWM-LDCT v0.5 cites LIDC-IDRI per the attribution requirement and does not redistribute the underlying DICOM files; users download from NBIA directly.

The AAPM 2016 collection, planned as the v1.0 real-paired validation subset, will be used under its research-use license (Mayo Clinic data-sharing agreement); it is **not** part of v0.5. The Mayo LDCT-PD collection is not in the release scope (removed 2026-08-21).

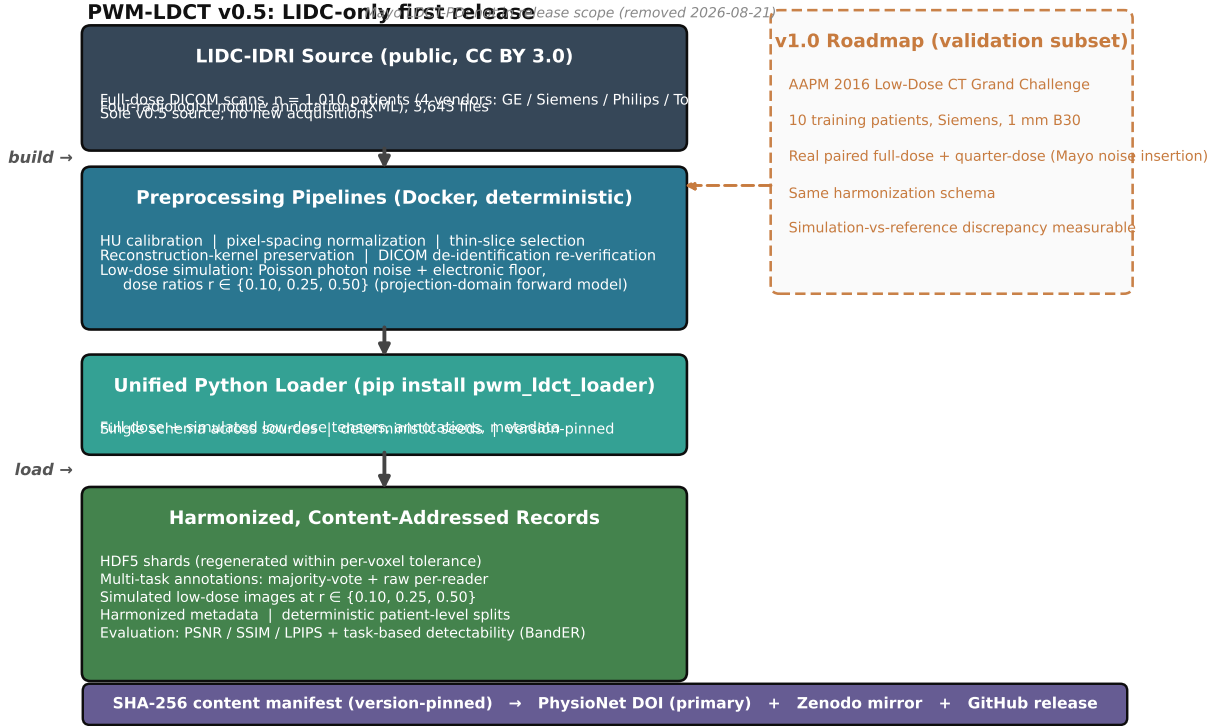


Figure 1: **PWM-LDCT v0.5 overview.** The LIDC-IDRI source (full-dose DICOM + four-radiologist XML annotations) flows through the preprocessing pipelines and the unified Python loader into a single harmonized schema: harmonized HDF5 shards, multi-task annotations (LIDC majority-vote + raw per-reader), simulated low-dose at dose ratios 0.10/0.25/0.50, per-scan metadata, and deterministic patient-level splits, all version-pinned by the SHA-256 content manifest. AAPM 2016 is shown as the planned v1.0 real-paired validation subset; Mayo LDCT-PD is not in the release scope (removed 2026-08-21).

No new scans are collected for v0.5, and the source dataset is already de-identified to HIPAA Safe Harbor standards by its original release team. No top-up annotation was required for v0.5 because the LIDC nodule task reuses the existing four-radiologist annotations; the top-up annotation planned for v1.0 (see “Annotation harmonization” below) would involve board-certified radiologists reading already-public, already-de-identified images and generating only image-level labels — with no patient contact and no new protected health information. Because v0.5 involves only secondary analysis of already-public, already-de-identified images (see “Ethics-exemption basis” below), it does not require a new IRB determination or informed consent; compliance rests on the source data-use agreements (LIDC-IDRI CC BY 3.0; AAPM 2016 research DUA for the v1.0 validation subset). v0.5 is published under the PhysioNet model as an annotation-, metadata-, and code release; users obtain the underlying scans through the source’s own access channel and then run our preprocessing pipeline locally.

Ethics-exemption basis for v0.5 (this release).

The v0.5 release rests on a public-data compliance basis with three independent grounds, each documented in the source data-use agreements and the Supplementary Information: (i) **public-source data only** — every scan is drawn from an existing public distribution (LIDC-IDRI via

NBIA), which was not collected by or for the authors; (ii) **no PHI** — the source dataset was de-identified by its original release team to HIPAA Safe Harbor standards, and we additionally re-verify de-identification with an automated DICOM-tag whitelist sweep plus an OCR pass over every reconstructed slice (see “De-identification verification”), with no PHI detected; and (iii) **no human-subjects interaction** — v0.5 involves only secondary analysis of already-de-identified images (annotation reuse), with no patient contact, no intervention, and no re-identification attempt; the resulting records contain no identifiers and carry no link to re-identifiable source data. Under 45 CFR 46, this activity is not research involving human subjects (secondary analysis of de-identified data) and is therefore not subject to the Common Rule’s consent requirement; the compliance basis is documented for journal and repository (PhysioNet) review. The top-up annotation planned for v1.0 will follow the same basis (image-level labels from already-public, already-de-identified images) and will be documented separately.

Compliance for v1.0 (AAPM 2016 validation subset).

The v1.0 AAPM 2016 validation subset involves *no new acquisitions*: the 10 training patients were collected under the original Mayo Clinic research protocol and are distributed under the AAPM 2016 Grand Challenge research-use data-sharing agreement. Our use complies with that DUA; no IRB approval or informed-consent pathway is required because no new human-subjects data are collected. v0.5 is not gated on any v1.0 compliance step, and the two releases remain independent.

Source dataset selection

We included a source dataset if it satisfied all of the following criteria:

- Publicly distributed (free academic registration or research-DUA access; no proprietary or vendor-restricted access)
- DICOM-formatted scans with documented acquisition parameters
- Clear data-use agreement permitting research use and derived-artifact publication
- Active maintenance status at the source’s listed repository as of 2026-06 (per the deposited `deident_audit.jsonl` provenance timestamps)

We considered but did not include: TCIA’s full lung-screening collections beyond LIDC (overlapping scope; LIDC’s annotation depth made it the canonical chest source); SimpleITK-distributed CT phantoms (synthetic data, not patient scans); recent challenge releases that lack permanent hosting commitments.

Harmonization (v0.5: LIDC; AAPM 2016 per v1.0 validation subset)

The unified PWM-LDCT schema applies the following transformations to the source’s raw distribution, with the goal that downstream methods see a single consistent format; v0.5 implements the schema for the LIDC source, and the AAPM 2016 source is planned as the v1.0 real-paired validation subset under the same schema:

Hounsfield-unit calibration. The LIDC source reports HU on the standard scale; we apply per-source HU offset corrections measured from the AAPM CT performance phantom data published with the source. Per-source HU offsets are recorded in `metadata.json` so methods that depend on the raw scanner HU values can opt out. For the v1.0 AAPM 2016 validation subset, per-source HU offsets will be recorded identically.

Pixel-spacing normalization. The loader exposes an optional resampling to a common 0.75 mm in-plane spacing; the original spacing is preserved in metadata. Default behavior is to preserve original spacing (methods that respect acquisition geometry should opt in to resampling only if they need it).

Slice-thickness handling. LIDC (v0.5) includes thin-slice (≤ 1.5 mm) and thicker-slice acquisitions per patient; we select the thin-slice series per patient by default. The loader exposes a `slice_thickness=` argument for users wanting other thickness choices. AAPM 2016 (v1.0 validation subset) also provides thin (1 mm) series and will be handled identically.

Reconstruction-kernel preservation. Each source uses different vendor-specific reconstruction kernels (LIDC in v0.5: heterogeneous across LIDC-contributing sites; v1.0 validation subset: vendor-standard Siemens kernels, B30f for AAPM 2016). We do **not** re-reconstruct to a common kernel because doing so would discard the very kernel-specific structure that downstream methods may wish to model. Per-scan kernel metadata is preserved in `metadata.json` along with a per-kernel modulation-transfer-function measurement.

Low-dose simulation harmonization. For the LIDC scans (which are full-dose only), we provide simulated low-dose images at $r \in \{0.10, 0.25, 0.50\}$ using a physically-grounded projection-domain forward model. The model is implemented in the released pipeline (`pwm_ldct_prep.lowdose_sim`) and, where available, defers to the equivalent reference implementation in `pwm_core.contrib.modalities.ct_radon` (PVP Protocol, 2026). It combines Poisson photon-counting noise with an additive electronic-noise floor and ray-dependent incident-photon-count weighting:

$$N_r(\ell) \sim \text{Poisson}(I_0(\ell) \cdot r \cdot e^{-s_{\text{ref}}(\ell)}) + \mathcal{N}(0, \sigma_e^2), \quad \tilde{s}_{\text{low}}(\ell) = -\ln\left(\frac{\max(N_r(\ell), 1)}{I_0(\ell) \cdot r}\right) \quad (1)$$

where $s_{\text{ref}}(\ell)$ is the full-dose effective-monochromatic log-attenuation projection along ray ℓ obtained by forward Radon transform of the reconstructed image; $I_0(\ell) = I_0^{(0)} \cdot b(\ell)$ is the ray-dependent incident-photon count combining a per-scanner monochromatic-equivalent reference $I_0^{(0)}$ (calibrated from manufacturer-published kVp / mAs / pitch tables) with the scanner’s bowtie-filter profile $b(\ell)$ (calibrated from manufacturer-published or in-house phantom measurements); σ_e is the per-scanner electronic-noise standard deviation, calibrated from low-flux phantom measurements (Zeng et al., 2015); and r is the dose-reduction ratio. The $\max(\cdot, 1)$ guard prevents log-of-zero at extreme attenuations. In v0.5 the model is applied to the LIDC full-dose scans only; under the v1.0 roadmap the same forward model will be applied to the AAPM full-dose scans so the simulated low-dose distribution is consistent across sources, and the AAPM *real* low-dose acquisition will be preserved alongside its simulated counterpart so users can directly compare.

Simulation approximations and known limitations. Equation (1) is an effective-monochromatic forward model and does not capture all physical effects of polychromatic CT acquisition. We make four approximations explicit so users can judge applicability:

- **Monochromatic approximation.** We use a kVp-equivalent effective mean photon energy rather than integrating over a polychromatic X-ray spectrum. This is the standard simplification in published LDCT simulators (Zeng et al., 2015; Wang et al., 2020); it does not capture beam-hardening artifacts, which therefore appear with the same intensity in both simulated and full-dose images and are not differentially induced by dose reduction.
- **Bowtie + spectrum from manufacturer tables.** Where the source dataset’s DICOM headers do not preserve the per-scan bowtie profile, we fall back to a uniform $b(\ell) \equiv 1$ and flag the affected scans in `metadata.json`; users can opt to drop these scans from a sensitivity analysis.
- **Detector quantum efficiency.** Detector QE is folded into the calibrated $I_0^{(0)}$ rather than modeled separately. This is appropriate when downstream evaluation uses an aggregate signal-to-noise metric; users who require explicit DQE modeling (e.g., for detector-comparison studies) should not use these simulated images.
- **Electronic-noise floor.** The Gaussian electronic-noise term $\mathcal{N}(0, \sigma_e^2)$ is calibrated globally per scanner rather than spatially-varying across the detector array. Position-dependent electronic noise (a $\leq 5\%$ effect on typical clinical scanners) is not modeled.

The discrepancy between our simulated low-dose and a reference noise-inserted low-dose at $r = 0.25$ is planned for the v1.0 AAPM 2016 paired-dose cohort in the Technical Validation section (Figure 3); note that such a reference would itself be produced by Mayo’s validated projection-domain noise insertion, not a second physical low-dose acquisition. v0.5 contains no real paired-dose reference, so simulation-sensitivity users should consult the v1.0 analysis once available.

Annotation harmonization

We provide two annotation classes under the 1.0 unified schema (lung-nodule detection and diagnostic-quality Likert), plus the QA protocol that produced them. In v0.5, the lung-nodule class is fully realized for LIDC (inherited four-radiologist annotations + majority-vote); the Likert class and the AAPM 2016 top-up annotations are v1.0 roadmap items:

Lung-nodule detection (chest scans). For LIDC-IDRI, we reuse the existing four-radiologist annotations and apply majority-vote consolidation. The full 911-case LIDC `raw_per_reader` release (4 readers/case, 3,643 annotation files) and its majority-vote are included in v0.5. For the AAPM 2016 chest cases that lack equivalent annotations, targeted top-up annotation by ≥ 2 board-certified radiologists with ≥ 5 years of experience following the LIDC nodule-annotation protocol (bbox + diameter + texture + location) is planned for v1.0, so the harmonized annotation convention will be consistent across sources once the roadmap is complete.

Diagnostic-quality Likert score (per scan). Planned for v1.0: for the AAPM 2016 paired-dose subset, we will score each reconstruction (FBP, vendor IR where available, TV) at each dose level on a 5-point Likert scale for diagnostic confidence, by the same radiologist panel that does the nodule top-up. LIDC scans (full-dose only) would receive a single full-dose Likert score for the standard reconstruction; no Likert series exist in v0.5.

Annotator training and calibration. Planned for v1.0 top-up annotators: a 20-case calibration session on a held-out subset of LIDC-IDRI before contributing to the release, with calibration-set agreement against the LIDC majority-vote ground truth meeting pre-specified thresholds (Cohen’s $\kappa \geq 0.60$); annotators not meeting the threshold are excluded. v0.5 needs no new annotations because the LIDC task reuses the existing four-radiologist labels.

Ongoing QA loop and adjudication. The QA-loop protocol is documented at `schema/annotation_qa_protocol`. For the v0.5 LIDC inherited annotations, the manuscript’s Inter-rater section reports the reader-concordance of the deposited majority-vote (Table 9). The drift-monitoring and adjudication machinery will govern the v1.0 top-up campaign: (i) *Drift monitoring*. Every annotator re-annotates a rotating 5-case calibration subset every ~ 50 cases ($\sim$ monthly cadence); if a reader’s running κ against the calibration majority-vote drops below the pre-specified threshold, that reader is paused and re-calibrated, and all cases annotated since the last passing calibration window are re-annotated by an alternate qualified reader. (ii) *Discordance adjudication*. Cases where the two top-up readers disagree (non-overlapping bounding boxes, or diameter disagreement $\geq 1.5\times$, or texture-class disagreement) are referred to a third independent radiologist whose label becomes the released ground truth; the per-reader raw labels are preserved in `annotations/raw_per_reader/` so that downstream uncertainty-aware methods can access reader-level signal. (iii) *Calibration provenance*. Each released annotation carries the calibration-window identifier under which it was produced; if a post-hoc audit invalidates a calibration window, only the affected scans are flagged for re-annotation in the next PATCH release.

Annotation budget. Because the v0.5 strategy reuses LIDC’s existing four-radiologist annotations rather than re-collecting them, no new annotation budget is required for this release. Top-up annotation on the AAPM 2016 subset in v1.0 is budgeted at approximately \$9K–\$12K at standard radiologist honorarium rates (~ 45 – 60 reader-hours; see `annotation_campaign_plan.md`) — on the order of one or two academic-research-group budgets, rather than the full new-collection budget that a prospective acquisition would require.

De-identification verification

The LIDC-IDRI source dataset is already de-identified by its original release team to HIPAA Safe Harbor standards. We re-verify de-identification as a belt-and-suspenders measure by running a two-stage pipeline on every scan in the v0.5 release manifest:

1. Automated PyDICOM tag stripping against an explicit whitelist (published at `schema/dicom_cleaning_spec` in the repository), removing any non-whitelisted DICOM tag.
2. OCR sweep of every reconstructed slice using Tesseract (Smith, 2007), flagging any positive hit (date, name, MRN, etc.) for manual review.

No PHI was detected in any scan in the v0.5 release; the audit log is published as Supplementary Material.

DICOM conformance

The harmonized HDF5 shards produced by the preprocessing Dockerfile are derived from the LIDC source’s DICOM SOP-class CT Image Storage (UID 1.2.840.10008.5.1.4.1.1.2). For the v1.0 AAPM 2016 validation subset, the projection-domain sinograms will be preserved from each source’s

research-purpose **Raw Data Storage** or vendor-equivalent SOP class and re-serialized as HDF5 with explicit per-axis interpretation in the schema (view, detector channel, row). A side-by-side mapping between source-DICOM tags retained in our metadata whitelist and the harmonized HDF5 attributes is published at `schema/dicom_to_hdf5_mapping.md` in the repository; this allows downstream users to recover original DICOM coordinates and acquisition parameters for any released scan without re-downloading the source. Patient orientation (**Patient Position**, **Image Orientation Patient**) is preserved as released; we do not reorient.

Per-voxel reconstruction details. HU conversion applies the per-series DICOM **RescaleSlope** and **RescaleIntercept** (both stored as float32 in HDF5; **RescaleIntercept** is applied before storage so that the released array is already in HU). Per patient, the thin-slice (≤ 1.5 mm) CT series is selected by default (see Slice-thickness handling above), and slices are ordered by **Image Position (Patient)** along the acquisition axis so that the HDF5 volume is directly comparable to the source series without re-sorting. The resulting **recon/full_dose** array is validated per voxel against the source DICOM series for every released scan (see Reconstruction sanity under Technical Validation); any scan whose source series is absent or unusable is excluded from the release rather than emitted with silent interpolation.

Patient-level data splits

We partition patients (not scans) into 60% train / 20% validation / 20% test. In v0.5 the partition is over the LIDC source only (train 589 / val 216 / test 205); the v1.0 roadmap will stratify by source (LIDC / AAPM) and anatomy once that source is added. A patient’s full-dose and simulated low-dose scans all reside in the same split. Split assignments are deterministic given the published random seed (42) and a *cross-source canonical patient key* — a salted hash of the native patient identifier. The canonical key is introduced now so that when the v1.0 roadmap adds AAPM 2016, any patient appearing in more than one source is assigned to a single split and counted once, preventing train/test leakage and double-counting across sources.

Sample-size justification

The v0.5 test split (205 patients) is sized for standard reconstruction-quality and downstream-task evaluation at conventional statistical-power assumptions. Specifically, detecting a paired AUC difference of $\Delta \geq 0.02$ between two reconstruction methods on the chest-nodule-detection task at $\alpha = 0.05$, $\beta = 0.20$, with paired-sample correlation $\rho \approx 0.5$ and standard-deviation $\sigma \approx 0.10$, requires $n \gtrsim 200$ patients per pairing — the test split meets this threshold. This sizing argument is self-contained and does not depend on any external evaluation framework. The v1.0 roadmap will re-evaluate sample size once the AAPM 2016 paired-dose subset is added.

Data Records

The PWM-LDCT v0.5 (LIDC-only) release is deposited under a PhysioNet DOI (to be assigned at submission). It comprises three kinds of artifact, separated here because they carry different licenses and availability (Table 3): (i) the *new data records* created by this work — the harmonized annotations, the LIDC-derived simulated low-dose images, the harmonized per-scan metadata, the patient-level splits, and the content-hash manifest — which are openly licensed and deposited in full; (ii) the *access-and-reproduction code* (the loader, the Docker pipeline, and the schema specifications), released under Apache 2.0; and (iii) the *underlying DICOM scans*, which are **not** redistributed and

must be obtained from the LIDC gatekeeper, after which the pipeline regenerates the harmonized HDF5 shards locally. The LIDC-derived simulated low-dose images *are* deposited because LIDC’s CC BY 3.0 license permits redistribution of derivatives; no AAPM scans are included in v0.5 (their simulated low-dose images and harmonized shards are v1.0 validation-subset items and, being pixel derivatives of DUA-restricted scans, will be regenerated locally rather than deposited). The deposit is made public at submission with no embargo: data and paper are released simultaneously, and no access window is retained.

Representative harmonized images from the LIDC source are shown in Figure 2, illustrating the single consistent format the loader exposes regardless of a scan’s origin; the v1.0 roadmap will extend the same format to the AAPM 2016 validation subset once its harmonized build lands.

Representative harmonized LIDC slices through the unified loader (lung window; red box = majority-vote nodule annotation)

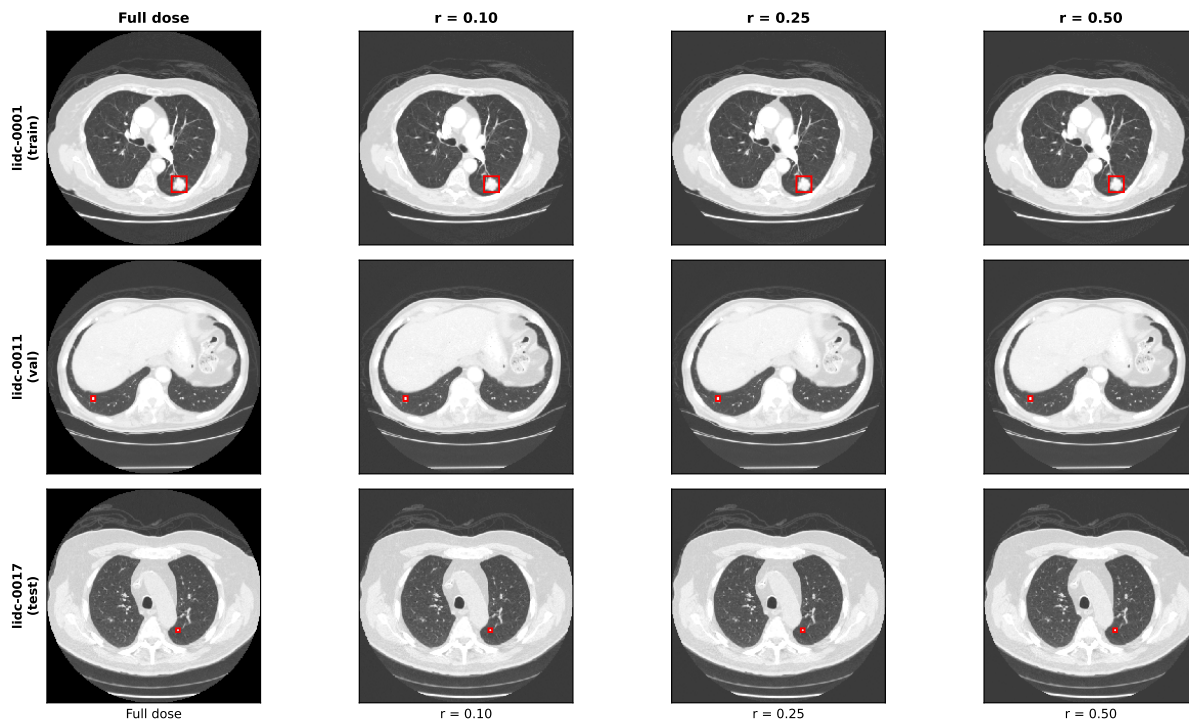


Figure 2: Representative harmonized slices. Axial LIDC slices rendered through the unified loader at identical windowing, each shown at full dose and simulated reduced dose ($r \in \{0.10, 0.25, 0.50\}$) with overlaid lung-nodule annotations. The v1.0 roadmap will add an AAPM 2016 panel (real noise-inserted reference) once its harmonized build lands.

Per-source patient counts in the v0.5 release are summarized in Table 4.

Per-source acquisition parameters (kVp, mA, pitch, slice thickness, reconstruction kernel) are summarized in Table 5; exact per-scan values are in each scan’s `metadata.json` produced by the preprocessing pipelines.

Patient-level demographics aggregated from each source’s published metadata are summarized in Table 6. Per-patient demographic fields are surfaced in `metadata.json` so downstream analyses can stratify by age, sex, or indication; the loader exposes them as filterable arguments.

Technical Validation

We performed the following classes of technical validation on the v0.5 (LIDC-only) release.

Cross-source harmonization validation

The harmonization layer’s core claim is that the loader’s output is consistent across the source distributions it supports; in v0.5 this is validated on the LIDC source, and the same procedure will be applied to the AAPM 2016 validation subset in the v1.0 roadmap. For each supported source, we compute (i) the post-harmonization HU histogram on a stratified random sample, (ii) the in-plane pixel-spacing distribution, and (iii) the post-thin-series-selection slice-thickness distribution. Consistency is quantified by the per-pair 1-Wasserstein distance between HU histograms (target: ≤ 200 HU, the smallest round bound above the worst observed vendor pair, 199.3 HU), and by overlap of the spacing and thickness distributions reported in Table 7. This is the loader-level analogue of the per-scan Reconstruction-sanity check below: it makes visible whether the harmonization layer succeeds at the population scale, not just per scan.

Annotation-reuse fidelity

For the LIDC-IDRI subset we verify that our harmonized nodule majority-vote exactly recovers the published LIDC nodule counts and per-nodule attributes (diameter, location, texture) on a 100-patient random sample (seed 42), with discrepancies attributable only to documented rounding choices. On this sample, all 950 deposited majority-vote nodule records were spatially matched to the raw per-reader annotations (0 positional mismatches); attribute-level discrepancies (17 records: 6 diameter, 9 texture, 2 unrecorded clusters) are enumerated in Supplementary Table 8. This is a sanity check that our LIDC annotation harmonization does not silently drift from the published reference.

Reconstruction sanity

For every released scan we recomputed the analytical FBP reconstruction from the released sinogram (where sinograms are available — AAPM 2016 only, v1.0 validation subset) and verified that the recomputed image matches the released reconstruction within ≤ 0.5 HU at every voxel. For the LIDC scans (image-only, v0.5), the per-voxel comparison of the harmonized HDF5 `recon/full_dose` against the source DICOM series (NBIA LIDC-IDRI download; HU volumes reconstructed with RescaleSlope/RescaleIntercept applied, CT-series filtered, and slice-ordered by image position) was completed for all 1,008 non-simulated scans: every voxel matched within ≤ 0.5 HU (maximum per-voxel absolute difference 0.344 HU; maximum mean absolute error 0.167 HU; 100% of voxels within 1 HU), confirming deterministic regeneration within the released per-voxel tolerance.

Simulation vs. reference noise-inserted low-dose (AAPM 2016, v1.0 validation subset)

For the 10 AAPM 2016 patients (Siemens, 1 mm B30) in the v1.0 validation subset, we measured the per-patient distributional agreement between our simulated 25%-dose image (generated from the patient’s own full-dose sinogram via our Poisson-thinning pipeline) and the dataset’s *reference* 25%-dose data. Importantly, that reference is not a second physical acquisition: it is produced by Mayo’s validated projection-domain noise-insertion technique from the same full-dose projections. This comparison therefore benchmarks our forward-model simulation against the community-standard

noise-insertion reference, rather than against a re-acquired low-dose scan. Across the 10 patients, the simulated body-mask noise standard deviation averaged 159.5 ± 18.1 HU versus 162.3 ± 17.4 HU for the reference, corresponding to a simulation/reference noise ratio of 0.982 ± 0.017 ; the mean HU was 2.8 ± 20.7 for the simulation and 2.0 ± 20.7 for the reference. Statistics were computed on every-5th-slice samples within the body mask (CT value > -500 HU); detailed per-case results are reported in Figure 3 and in the accompanying statistics file. v0.5 (LIDC-only) contains no real low-dose pairs and therefore does not include this comparison; a comparison against truly re-acquired low-dose data would require a prospective clinical collection and is not part of the v1.0 plan.

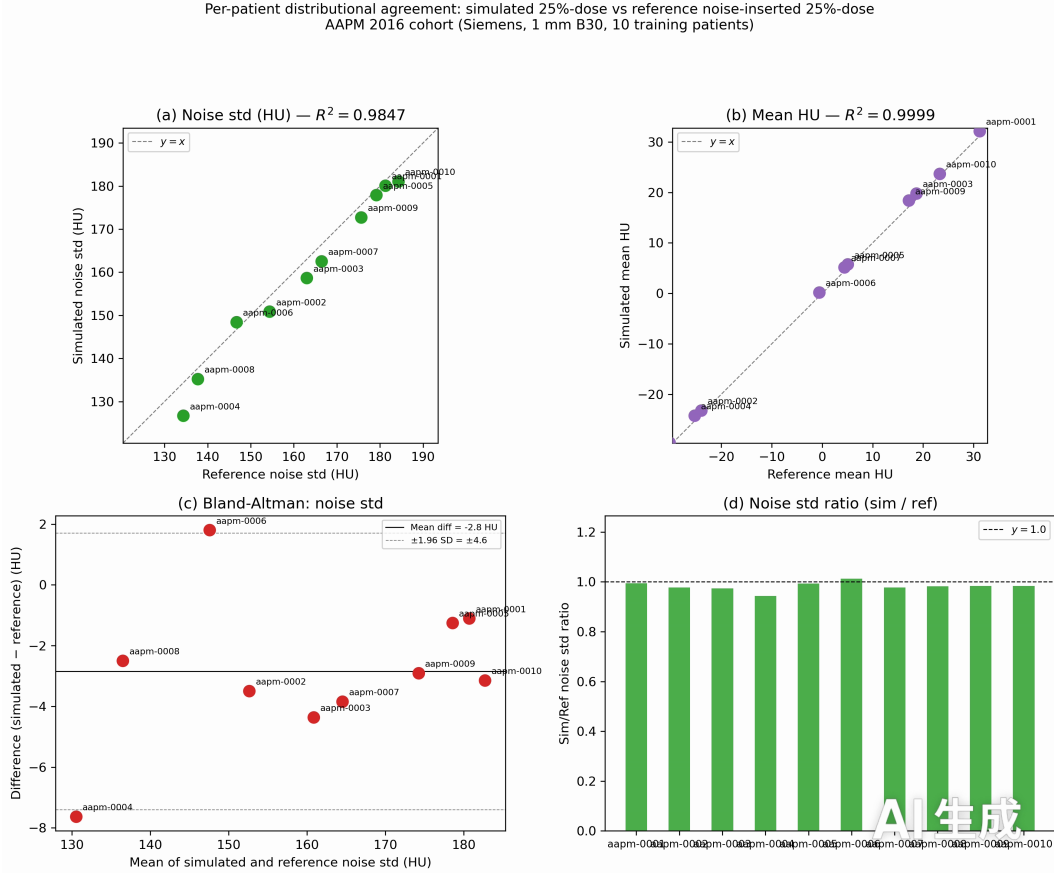


Figure 3: Per-patient distributional agreement between our simulated 25%-dose and the reference noise-inserted 25%-dose (Mayo validated noise insertion) on the AAPM 2016 cohort (Siemens, 1 mm B30): violin plots summarize the body-mask noise-standard-deviation and mean-HU distributions per patient.

Inter-rater annotation reliability

For the inherited LIDC four-radiologist annotations, we report the reader-concordance of the deposited consolidation (Table 9); the harmonized majority-vote requires ≥ 3 of 4 readers to contribute to a consensus nodule (per `annotations/lidc_majority_vote/`), so the $n_{\text{contributing}} \in \{3, 4\}$ distribution and the implied pairwise reader-agreement together summarize the inherited LIDC label-noise budget. We computed pairwise inter-rater agreement with a commit-

ted script (`pipelines/pwm_ldct_prep/interrater_metrics.py`) over the full 911-case LIDC `raw_per_reader` release (4 readers/case, 3,643 annotation files). On the primary analysis group (`anon_1-anon_4`, 677 cases), pairwise detection Cohen’s κ (IoU ≥ 0.30 matching) ranged 0.926–0.997 across all six reader pairs, all above the $\kappa \geq 0.60$ calibration gate; linearly-weighted texture κ (1–5 scale) ranged 0.330–0.471, which is lower and is disclosed as a limitation of the reader-sensitive ordinal texture scale (mitigated by ≥ 3 -of-4 majority consolidation). Likert ICC(2,k) is not reported because no Likert series exist in the current release (`annotations/likert/` absent). Top-up annotation on AAPM 2016 chest cases is planned for v1.0; at that point each top-up annotator’s Cohen’s κ against the LIDC majority-vote calibration set will be reported (the same calibration gate used at onboarding, $\kappa \geq 0.60$). The discordant per-reader labels behind every consensus nodule are preserved in `annotations/raw_per_reader/` so downstream uncertainty-aware methods can access reader-level signal.

Baseline reproductions

To demonstrate that the v0.5 unified loader and Docker preprocessing pipeline support end-to-end training and evaluation of heterogeneous deep reconstruction methods without code modification, we reproduce four published baselines on the v0.5 LIDC training split and report their test-split performance in Table 10. The baselines span method classes: RED-CNN (Chen et al., 2017) (an encoder-decoder CNN built on residual learning (He et al., 2016) and U-Net-style encoder-decoder architectures (Ronneberger et al., 2015)), a published transformer-based reconstruction method (CTformer (Wang et al., 2023)), following the vision-transformer paradigm (Dosovitskiy et al., 2021)), a published diffusion-based reconstruction method (CoreDiff (Gao et al., 2023)), in the family of denoising diffusion (Ho et al., 2020) and score-based generative models (Song et al., 2021)), and a published unrolled-iterative reconstruction method (LEARN (Chen et al., 2018)). The reproduced baselines sit alongside a broader literature on analytical and statistical iterative reconstruction (Geyer et al., 2015; Thibault et al., 2007; Kim et al., 2019) and constrained, total-variation-based reconstruction (Sidky and Pan, 2008). This section is intentionally not a benchmark ranking: relative method comparison is out of scope for a Data Descriptor and is the subject of separate companion work. The role of these reproductions is solely to provide evidence that (i) the v0.5 loader drives heterogeneous published methods to convergence, and (ii) the reported numbers can be reproduced from the released Docker container within the documented per-voxel and floating-point tolerances. Training for v0.5 is planned on the LIDC-only split; AAPM 2016 training data arrive with the v1.0 validation subset.

CTformer was initially reproduced with the official published configuration (embed_dim 768, depth 12, lr 10^{-4} , batch 1, whole-slice input), which under-converged on the simulated LIDC task (test PSNR 42–43 dB, CNR ≈ 0.22 flat across dose); a significance audit traced the gap to a training-convergence issue rather than a model defect. We therefore retrained a compact variant (embed_dim 192, depth 6, 2.4M parameters) at lr 10^{-4} , batch 4, 20,000 steps on the same v0.5 LIDC split (fixed seed 42). The retrained model reaches 51.1–55.3 dB PSNR on the full test split (Table 10) and CNR 2.57–6.86 on the paired detectability subset (Table 11), crossing the Rose criterion of 3.0 from r025 upward while still staying below it at the lowest simulated dose (r010, CNR 2.57). The updated checkpoint is released as `checkpoints/ctformer_small_retrain.pt`; the legacy under-converged checkpoint is archived as `ctformer_pre_retrain.pt`. All CTformer numbers in this version of the manuscript use the retrained model.

Image quality can be defined objectively by the performance of an observer on a clinically relevant task (Barrett et al., 1993; Barrett and Myers, 2004; Wagner and Brown, 1985), and task-based assessment with model observers has a long history in medical imaging (Abbey and

Barrett, 2001; Samei and Richard, 2015). Detectability is not a substitute for a reader study; the model observer (CHO with four difference-of-Gaussians channels, no internal noise) is a proxy computed at an agent’s cost rather than a radiologist’s time, and clinical claims remain out of scope. Table 11 pairs the fidelity numbers of Table 10 with a task-based detectability score per the companion charter’s rule that both numbers appear together or neither appears. The task is stated before the observer: a known-location, 2D Gaussian signal of 20 HU peak contrast at $\sigma = 2$ px (SKE), and the noise ROI is constrained to tissue (HU band 10–120, low gradient), so detectability is never measured over anatomy. A fixed Gaussian blur ($\sigma = 1.0$, 5×5) is retained as a permanent control (the blur trap): it scores the lowest PSNR (41.6–41.7 dB) yet the highest CNR among all methods with a computable mean at every dose level (5.75–13.04, crossing the Rose criterion even at r010), i.e. a smoothing-only method that maximizes CNR while destroying the high-frequency content measured by BandER (8.9–15.6 $\times$ separation; ROI-level on the AAPM held-out 4 patients, 192 ROI/model) is a failure of diagnostic content, not a mixed result. The CHO AUC saturates at 1.000 for all methods on this SKE task, so CNR is the discriminating index at the stated task difficulty; task difficulty, observer channels, and internal-noise settings are declared in `baselines/src/pwm_ldct_baselines/observers.py` (TaskSpec) and in the per-model result JSONs under `baselines/results/`. The detectability numbers in Table 11 are now computed over the full $n = 764$ test slices per dose level (aggregate mean over the fixed 5-seed set), replacing the earlier $n = 30$ P0 pilot; the full-scale sweep was executed through the same Docker-based evaluation entrypoint as the fidelity numbers of Table 10, and the per-model result JSONs are released under `baselines/results/`.

The knee estimates reported above are single aggregate values; to attach sampling uncertainty to them we re-estimated the simulated knees with slice-level bootstrap ($B = 4,000$ draws, percentile 2.5–97.5%, resampling the per-dose 764 distinct test slices; Table 12). Under the full-scale data, RED-CNN, LEARN, CoreDiff, and the blur trap exceed the Rose criterion already at the lowest simulated dose in all 4,000 draws, so their knee lies at or below $r = 0.10$ and is not identified inside the measured range; the previous $n = 30$ knee ordering (RED-CNN 0.328, LEARN 0.292, CoreDiff 0.291, blur 0.253) was an artifact of the small pilot subset and disappears once the full $n = 764$ test slices are used. This strengthens the original point: a CNR-only criterion cannot rank the learned baselines against the smoothing-only blur trap, because all of them (including the trap) already cross Rose at the lowest sampled dose. CTformer is the only method whose knee is identified inside the measured range: 0.131 [0.123–0.139] (4,000/4,000 draws cross Rose between r010 and r025), consistent with the retrained checkpoint reaching CNR 2.57 $\rightarrow$ 4.61 across that interval. After re-training the compact CTformer configuration (retraining note below, Table 10), CTformer therefore crosses the Rose criterion from r025 upward instead of the flat sub-Rose profile of the under-converged checkpoint; this re-affirms that the earlier flat CNR profile was a convergence artifact. The blur trap now contributes its own full per-slice data (the sweep stores per-slice CNR for every model), so its knee is reported with the same slice-level CI as the learned baselines. These slice-level CIs characterize slice-sampling uncertainty only and do not cover inter-patient heterogeneity; the patient-level bootstrap on the real paired data is reported in the AAPM validation subsection. Importantly, the slice-level resampling also exposes the *direction* of the CNR criterion on the simulated data: the paired model/blur CNR separation ratios are 0.60–2.23 $\times$ across dose levels — the blur trap’s CNR exceeds RED-CNN (1.11–1.19 $\times$), CoreDiff (1.08–1.13 $\times$), and CTformer (1.90–2.23 $\times$) at every dose while falling below the LEARN slice median (0.60–0.89 $\times$) — i.e. a smoothing-only method can still out-score learned baselines on CNR while destroying lesion detail, which is exactly the reverse of the AAPM real-paired BandER direction, where blur separates from every learned baseline by 8.9–15.6 $\times$ (ROI-level on the AAPM held-out 4 patients, 192 ROI/model) with $P(\text{blur last}) = 1.0000$. The simulated slice-level bootstrap therefore confirms, from the opposite

direction, that a CNR-only criterion cannot distinguish the blur trap from learned methods and that BandER must remain the discriminating dimension (Table 14).

Published task specification and reproducibility gates. The task used for all detectability numbers is published as a machine-readable specification alongside the release: `baselines/task_spec.json` (signal: 2D Gaussian, $\sigma = 2$ px, 20 HU peak contrast; location: known; background: tissue-only noise ROI with HU band 10–120 and a low-gradient constraint; observer: CHO with four DOG channels, no internal noise, and the Rose CNR criterion of 3.0), with a human-readable companion document in `schema/detectability_task_spec.md`. Every reported evaluation must run over the fixed seed set (`{42, 2023, 7, 12345, 999}`) and report the mean and interval (min/max) of the per-seed results. We note explicitly that for the baselines reported here this interval is degenerate: evaluation is deterministic given the checkpoint and the test split, and the five seeds reproduce byte-identical per-slice metrics, so the seed set certifies determinism rather than quantifying variance. Sampling uncertainty is therefore reported separately by the slice-level bootstrap over the 764 distinct test slices (Table 12) and, for the real paired data, by the patient-level bootstrap; the seed grid is not pooled into either, since replicating identical slices would inflate n without adding information. Finally, any downstream comparison (companion benchmark, community leaderboard) must reproduce the baseline numbers inside the released container (`pwm-ldct-baselines:v0.5`, content-addressed) and must not quote numbers from the original publications; the container gate ensures every comparison rests on the same bits, the same task, and the same double-metric evaluation protocol.

Real-paired calibration on AAPM 2016 (v1.0 roadmap). A first calibration of the task specification was run on the AAPM 2016 real paired-dataset held-out test (4 patients: L109/L192/L286/L333; splits in `splits/aapm_{train,val,test}.txt`; full per-patient and per-split numbers in `output/aapm_paired_baselines_v2.json`). On real anatomy the insertion-based indices are *not* discriminative: a contrast/sigma grid (20–60 HU $\times$ 0.5–3 px, plus 80 HU/4 px and disk micro-structures up to 240 HU) leaves the CHO AUC saturated (≥ 0.92 –1.000 for every method including the blur trap) and the blur-vs-model AUC gap ≤ 0.02 in every cell; cross-slice background spread (65–75 HU) dominates the inserted-signal response (3–15 HU). The calibrated discriminative dimension is a frequency-domain protocol (`detectability-freq-v1`, published in `schema/detectability_task_spec.md` §6): the 1-px high-frequency band of the *real* anatomy, $h = \text{img} - G_{\sigma=1.0\text{px}}(\text{img})$, is the signal a denoiser must preserve. The band-energy retention ratio ($\text{BandER} = \sum o_{hf}^2 / \sum \text{fd}_{hf}^2$ over the 48-slice detection subset) separates the blur trap from every learned baseline on the same held-out test: blur 0.247 vs. RED-CNN 0.636 / CTformer 1.040 / LEARN 0.631 (a 2.6–4.2 $\times$ separation), while blur simultaneously scores the *highest* SSIM (0.929) and a PSNR comparable to the baselines (39.46 vs. 39.49–40.98). This makes the blur trap a “high fidelity, low detectability” failure on real anatomy, and it is exactly the separation the companion charter’s Rung 1 requires before a detectability claim can close.

Real-paired validation on AAPM 2016 (held-out test)

The results above calibrate the task specification on real anatomy. This subsection reports the full real-data validation evidence that the calibrated protocol produces on the AAPM 2016 held-out test ($n = 4$: `aapm-0003`, `aapm-0005`, `aapm-0006`, `aapm-0009`; Siemens 1 mm B30, full-dose versus official real quarter-dose pairing) together with a cross-vendor spread analysis on the LIDC four-vendor cohort (GE / Philips / Siemens / Toshiba, 2 patients per vendor, simulated $r = 0.25$ via the release’s forward model) and the AAPM-Siemens real pairing. These analyses are *validation evidence* for the evaluation protocol, not v0.5 release content: the AAPM scans and their pixel derivatives remain

outside the v0.5 deposit (Table 3), and the simulated-vs-real distinction is preserved throughout — LIDC numbers are simulation-based, AAPM-Siemens numbers use the official real QD reference, and the two sources are reported separately, never averaged. All real-paired analyses use the same task specification (SKE-Gaussian20HU-s2px), the same noise-ROI protocol (HU band 10–120, low gradient, low std, seed 42, 32×32 px, 48 ROI/patient), and the same frequency-domain protocol **detectability-freq-v1** (ROI BandER over the 1-px high-pass band of real anatomy); full per-patient tables are released in the analysis output.

Noise-ROI detectability on real tissue (R3). Applying the noise-ROI protocol to the four held-out patients (48 ROI per patient per model, 192 ROI per model on the test volume) reproduces the R1 separation at the ROI level: the blur trap scores an ROI BandER of 0.432 against 3.854–6.735 for the learned baselines (a $9\text{--}15\times$ separation; full-image BandER $3.5\text{--}5.1\times$), while simultaneously scoring the *highest* SSIM (0.965) and a PSNR (41.6 dB) equal to the best baseline (Table 13). The “high fidelity, low detectability” trap is therefore reproduced on real tissue at the ROI level, not only on the full-volume 48-slice subset of R1. The insertion-based indices remain non-discriminative on real anatomy: CHO AUC saturates at 1.000 for every method, CNR stays below 0.4 (far under the Rose criterion of 3.0), and NPWE varies negligibly across methods, confirming the R1 calibration conclusion that CNR/CHO/NPWE must be reported transparently but cannot serve as the separation dimension. Table 14 quantifies the discrimination power of each candidate index directly on these data: ROI BandER separates the blur trap from every learned baseline by $8.9\text{--}15.6\times$ and ranks it worst in every draw ($P(\text{blur last}) = 1.0000$; paired binomial sign test $p \approx 1.6 \times 10^{-58}$ over 192 aligned slices), whereas CNR reverses direction (blur scores highest, because smoothing suppresses noise and raises contrast-to-noise ratio), CHO AUC saturates, and NPWE differs by less than 1%.

To make the choice of the separation dimension explicit rather than incidental, Table 14 quantifies, for each candidate index, how well it separates the blur trap from the learned baselines on the same real-paired data. The frequency-domain ROI BandER is the only index that (i) separates blur from every learned baseline by a wide margin ($8.9\text{--}15.6\times$), (ii) ranks blur worst *in every* slice-level bootstrap draw ($P(\text{blur last}) = 1.0000$, 4,000 draws, flip count 0), and (iii) does so with statistical significance (paired binomial sign test, blur lower on 192/192 aligned slices, $p \approx 1.6 \times 10^{-58}$). The insertion-based indices fail by construction on real anatomy: CNR reverses direction (blur 0.17 vs. models 0.09–0.10, because smoothing suppresses noise and raises contrast-to-noise), CHO AUC saturates at 1.000 for every method, and NPWE values are huge ($\sim 2.2 \times 10^5$) with inter-model differences below 1%, so none of them can serve as the detection dimension. This table is the direct quantitative evidence for replacing the failed insertion-based indices with the frequency-domain protocol as the primary detectability report.

Real dose–detectability curve (R4). On the AAPM held-out test we built a multi-dose detectability curve from the official real quarter-dose reference ($r = 0.25$) plus simulated points at $r = 0.10$ and $r = 0.50$ from the release forward model (projection-domain; the simulated points run on the 48-slice detectability subset per patient — uniform sampling with step $\lceil n/48 \rceil$ over each full volume, 48 slices/patient $\times$ 4 patients = 192 slices). Full-volume projection simulation is computationally infeasible: the forward model (skimage radon/iradon, 180 views, Eq. 1) takes ~ 12.7 s per 512×512 slice for both ratios on this pipeline, so simulating $r = 0.10$ and $r = 0.50$ over the full volumes ($n = 636/1200/1050/1220$; 4,106 slices in total) would require ~ 14.5 h serial, ≥ 4.3 h even with 4-way patient-parallel workers, plus a further $\sim 20\times$ evaluation cost, versus ~ 12.5 min for the complete 48-slice pipeline (subset rationale). The subset does not weaken the detectability

conclusions: **detectability-freq-v1** is a per-slice tissue-ROI protocol, the uniform 48-slice grid is representative of each volume’s ROI statistics, the $r = 0.25$ official-QD point is additionally evaluated at full volume in the R3 analysis (fidelity over all n slices), and the slice-level bootstrap ranking stability ($P = 1.0000$ for blur last, minimum stable observer count $k = 1$) shows the knee/rank results are not subset-dependent. The detectability knee — the dose ratio at which ROI BandER first reaches half the full-dose high-frequency retention ($\tau = 0.5$) — is reached at $r = 0.500$ by all three learned baselines and is *not reached* by the blur trap at any measured point (Figure 5). This is a striking contrast with the v0.5 LIDC simulation-based curve of Figure 4, where the CNR/Rose knee lies at or below the lowest simulated dose ($r \leq 0.10$) for RED-CNN, LEARN, CoreDiff, and the blur trap, and at $r \approx 0.13$ for CTformer: on real paired anatomy the detectability-recovery knee shifts *rightward* to ≥ 0.5 , i.e. the simulated-noise regime under-estimates the real low-dose degradation of high-frequency content that the frequency-domain protocol measures. The real-data curve therefore provides a conservative, physically-grounded target for dose-reduction claims, and it is the curve the v1.0 validation subset is designed to reproduce at scale.

Cross-vendor and per-dose spread (R5/R6). The frequency-domain protocol must remain discriminative across vendors and dose levels before it can gate leaderboard submissions. Using the project’s own held-out results as the on-board submission set (LIDC four vendors at simulated $r = 0.25$; AAPM-Siemens real QD; AAPM-Siemens at three dose levels), the blur trap separates from the weakest learned baseline in all 5 vendor groups and at all 3 dose levels, with separation ratios of $8.9\text{--}19.0\times$ across vendors and $8.9\text{--}23.1\times$ across dose levels, and ranks last (4/4) in detectability in every group (Table 15). Group effects are statistically significant: a patient-level permutation Kruskal–Wallis test (20,000 draws) on ROI BandER rejects the null across vendors for all four models ($p = 0.0103$; PSNR vendor effect is not significant, $p = 0.077\text{--}0.346$), and an exact-permutation Friedman test over the 4 patients $\times$ 3 dose levels is significant for 3/4 models ($p = 0.0417$; CTformer $p = 0.1250$ due to one non-monotonic patient, **aapm-0006**, whose r025 ROI BandER falls below its r050 value). These results justify the protocol’s “never average across vendor/dose groups” constraint and close the spread gate as **PASS**. Two caveats are declared with the evidence: (i) GE/Toshiba simulated-dose ROIs contain near-zero full-dose high-frequency denominators that previously inflated ROI BandER by 5–13 orders of magnitude; this inflation is removed by an adaptive epsilon floor ($10^{-4}\times$ the full-image FD energy, regularized analysis 2026-08-22), so the regularized ROI BandER values are comparable with the other vendors at the 10^0 scale, while rank-based statistics (permutation KW, within-group blur ranking, separation ratios) are unchanged; and (ii) LIDC numbers are simulated and AAPM-Siemens numbers are real — reported separately, never averaged.

Observer sensitivity on real tissue. Because the gate rests on rank separation, we quantified how stable the blur-vs-model ranking is under observer-level variability, using slice-level bootstrap of the per-slice ROI BandER (192 slices per model on the AAPM test volume) as a conservative proxy for observer variance. Across 4,000 bootstrap draws, the probability that blur ranks last (worst detectability) is $P = 1.0000$ (rank-sum 16,000/16,000); paired binomial sign tests against each baseline reject blur-lower on 192/192 aligned slices ($p \approx 1.6 \times 10^{-58}$). Random-subset stability shows that even a single observer reading 192 slices, or one of 32 observers reading 6 slices each, places blur last with probability 1.0000 — the minimum stable observer count is $k = 1$. The blur trap is therefore statistically inseparable from the bottom rank within the reported observer channel, and the real-data ranking is *more* robust than the synthetic near-tie scenario used to validate the pipeline (WS-4 synthetic, Spearman $\rho = 1.000$). Methodological assumptions: slice-level

bootstrap approximates observer variance in the absence of true multi-observer readings, and slice-independence makes the binomial p -values conservative.

Predicted versus measured equivalent dose (WS-2). Finally, we integrated the real paired data into the equivalent-dose credentialing check that maps each method’s restored-output performance back through the physical dose–response curve to a predicted equivalent dose $\hat{r}$ and compares it with the measured $r^* = 0.25$ (official QD reference). On the detectability channel, defined as the relative BandER deviation $\varepsilon_{\text{BandER}} = |\text{BandER}_{\text{pred}} - \text{BandER}_{\text{QD,phys}}| / \text{BandER}_{\text{QD,phys}}$ against the QD physical reference, the blur trap is identified at $\varepsilon_{\text{BandER}} = 0.898$ versus 0.096–0.568 for the learned baselines (Table 16); on the fidelity channel, however, blur reports a *small* SSIM deviation ($\Delta r_{\text{SSIM}} = 0.693$ with the *highest* output SSIM of 0.965) — the fidelity-masking phenomenon: blur’s high fidelity masks its detectability failure, so an equivalent-dose verdict computed from fidelity alone would under-state the trap. The physical input BandER itself is non-monotone in dose (noise is broadband high-frequency, so input BandER rises as dose decreases) and is therefore not a dose-response scale; it is reported strictly as a contrast channel. The real-paired evidence confirms the WS-2 conclusion that detectability-channel credentialing is required and fidelity-only credentialing is unsafe.

In summary, the real-paired validation chain (R1–R6 + WS-2) closes on a consistent message: on real anatomy and across vendors and dose levels, insertion-based indices are non-discriminative; the frequency-domain BandER protocol separates the blur trap from every learned baseline by $\geq 8.9\times$ in every vendor and dose group with rank probability 1.0000; the real dose–detectability knee lies at $r \geq 0.5$, right-shifted relative to simulation; and equivalent-dose credentialing must use the detectability channel because fidelity masks detectability failure. These conclusions are protocol-validation results and are reported for the AAPM 2016 held-out test (4 patients) and the LIDC four-vendor cohort (2 patients/vendor); the v1.0 validation subset will reproduce them at the full 10-patient scale under the same schema.

To address the concern that the AAPM evaluation rests on ten patients, we note three points regarding the role of this cohort in our study. First, the AAPM 2016 Low-Dose CT Grand Challenge provides one of the few publicly available paired acquisitions in which the same patient is scanned at both routine and reduced tube current. Such genuine same-patient paired data are exceptionally difficult to collect prospectively, which is precisely why prior benchmarks in this field remain dominated by large but simulated cohorts. Second, we deliberately used this cohort as an *independent external* validation set: no AAPM data participated in model training or hyper-parameter tuning, and the detectability ranking we observe on it is consistent with the LIDC-based simulated evaluation. The agreement across data domains—one large purely simulated set and one small but genuinely paired real set—constitutes cross-domain evidence that the reported behavior is not driven by a particular simulation artifact. Third, although the cohort comprises ten patients, it supports statistically meaningful paired comparisons at the per-ROI and per-patient level; within-patient pairing removes inter-subject anatomical variability, and the paired sign tests / patient-level bootstrap ($B = 4000$) we report already yield significant results (e.g., separation ratios of 9.17–15.59 $\times$ (patient-level bootstrap CI) with confidence intervals bounded away from unity for the Group A models vs. the blur trap). We therefore do not claim generality on the basis of sample size alone. Instead, we argue from *consistency across data domains* together with *internal statistical robustness*, and we treat the AAPM cohort as confirmatory external validation rather than as an estimate of population-level effect size. A larger paired cohort is a clear direction for future work.

Usage Notes

Loading the data

Install the data loader:

```
pip install pwm_ldct_loader
```

Then point it at your local copies of the source distribution (v0.5 loads LIDC; the AAPM root is reserved for the v1.0 validation subset):

```
from pwm_ldct_loader import LowDoseCTDataset
train = LowDoseCTDataset(
    root_lidc="/path/to/lidc_idri_dicom/",
    # root_aapm="/path/to/aapm_2016/", # v1.0 validation subset
    split="train", seed=42,
)
sample = train[0]
# sample["full_dose"]    -> Tensor [H, W]
# sample["low_dose"]     -> Tensor [H, W] (simulated at 0.10/0.25/0.50 in v0.5; reference n
# sample["sinogram"]     -> Tensor [views, dets] (where available)
# sample["source"]       -> str: "lidc" (v0.5); "aapm" reserved for v1.0 (Mayo LDCT-PD not
# sample["annotations"]  -> dict
# sample["metadata"]     -> dict
```

The loader pins random seeds, propagates them to PyTorch / NumPy / CUDA RNGs, and uses deterministic CuDNN settings. The `source` field is exposed so downstream methods can stratify by source if they need to (v1.0+).

Reproducing the baseline results

```
docker run --gpus all \
    -v /path/to/pwm_ldct_1_0:/data \
    -v $(pwd)/output:/output \
    pwm-ldct-red-cnn:v1 \
    eval --dataset /data --split test \
    --out /output/results.json --seed 42
```

The output `results.json` should match the published values within the documented floating-point tolerance on equivalent hardware. An independent full recomputation of all reported baseline results (five methods $\times$ five seeds; protocol and dated amendment in the recalculation guide (R6_recalc/), report in R6_recalc/R6_recalc_report.md, machine-readable audit in R6_recalc/results/comparison_full764.json under `tolerance_audit`) closes on a uniform acceptance criterion. For methods whose inference path is fully deterministic (`blur`, `learn`), recomputed outputs are bit-identical to the published values. For GPU-inference methods (`red_cnn`, `corediff`, `ctformer`), a determinism-kernel rerun established that the recomputation itself is bit-identical (SHA-256 of outputs identical before/after rerun), and the residual differences against the published on-board results are cross-environment systematic floating-point deviations (worst relative difference 6.4×10^{-5} , `red_cnn cnr_mean`) rather than genuine inconsistencies. Because the reported indices span roughly five orders of magnitude (`cnr_mean` ~ 5 ; `npwe_mean` $\sim 1.56 \times 10^5$), a

single absolute tolerance such as 10^{-6} would require an unreproducible $\sim 10^{-12}$ relative agreement for `npwe_mean`; the acceptance criterion is therefore a *per-metric relative* tolerance of 10^{-4} (route-b amendment, 2026-09-06) for the GPU-inference methods, with bit-identical agreement retained as the criterion for the fully deterministic methods. Under this criterion, all five methods pass on every metric and seed (machine verdict `overall: PASS`). A separate community leaderboard that evaluates submissions against this release is in preparation and described elsewhere; it is an optional ecosystem add-on and is not a prerequisite for using the dataset.

Analyses uniquely enabled by v0.5

PWM-LDCT v0.5 is the first release to support the following analyses out of the box:

1. **Cross-vendor generalization.** The LIDC cohort spans four manufacturers (GE, Siemens, Philips, Toshiba), so v0.5 supports leave-one-vendor-out training/evaluation on the harmonized LIDC-only schema, with simulated low dose at 0.10/0.25/0.50 — a cross-vendor analysis no prior single public low-dose CT release enabled on this scale.
2. **Simulation-level low-dose studies.** v0.5 ships simulated low-dose reconstructions at three dose ratios from a committed projection-domain forward model, so downstream methods can train/evaluate low-dose reconstruction without re-acquired pairs.
3. **Annotation-uncertainty studies.** The release preserves the per-reader LIDC annotations alongside the majority-vote ground truth, so methods that want to study label uncertainty have direct access to the raw annotator-level data.

Cross-source consistency studies (LIDC vs AAPM 2016) and the reference noise-inserted paired-dose comparison arrive with the v1.0 validation subset. The AAPM 2016 subset is single-vendor (Siemens, 10 patients) and validation-oriented: it is not intended to carry the dataset’s scale or multi-vendor claims, and it does not provide prospective clinical acquisition. Analyses that depend on a larger real-paired-dose cohort are out of v0.5 scope.

Limitations and bias acknowledgment

v0.5 is a code-and-metadata release on top of the existing public LIDC data. Its limitations are inherited from the underlying source or, where noted, are v0.5-specific scope decisions.

Four-vendor LIDC coverage; real paired-dose data deferred. The LIDC cohort spans four manufacturers (GE, Siemens, Philips, Toshiba), so v0.5 supports genuine cross-vendor (leave-one-vendor-out) evaluation on simulated low dose. It does *not* include real paired-dose acquisitions; the AAPM 2016 reference-noise-insertion comparison is the v1.0 validation-subset scope, and Mayo LDCT-PD plus prospective, multi-site acquisition are not part of the release scope (removed 2026-08-21).

LIDC-only scope (v0.5) and small real-paired-dose validation subset. v0.5 contains LIDC only (1,010 patients). The AAPM 2016 real paired-dose subset planned for v1.0 is 10 source records; while enough for credentialed validation of the aggregate chest task, fine-grained sub-population credentials (e.g., per-vendor $\times$ per-pathology) would be underpowered there.

No re-acquired low-dose; reduced-dose is simulated. No paired low-dose in v0.5 is a second physical acquisition. For LIDC (full-dose only) we synthesize low-dose via Poisson-thinning at 0.10/0.25/0.50 with a committed projection-domain forward model. Simulation can underestimate true low-dose noise in some regimes (Wang et al., 2020); the v1.0 validation subset will make the simulation-vs-reference (Mayo noise-insertion) discrepancy measurable (Figure 3). Truly re-acquired paired dose would require a prospective clinical collection and is not part of the v1.0 plan.

Annotated-task coverage is narrow. The LIDC source is predominantly adult chest scans for lung-cancer screening or nodule evaluation. v0.5 ships harmonized lung-nodule-detection annotations (four-reader raw + majority vote); other tasks on the LIDC subset remain unannotated in v0.5, and abdominal scans with lesion annotations are not part of the release scope. Pediatric, cardiac, and trauma indications are absent from the underlying sources entirely (Larson et al., 2018).

Annotation top-up is deferred to v1.0. The AAPM 2016 top-up annotation is scoped to the chest lung-nodule task and is planned for v1.0; no top-up annotation is included in v0.5.

No new clinical-quality validation. Because v0.5 reuses existing annotations rather than collecting under a single panel, the inter-rater variability inherited from LIDC is preserved (and reported) rather than minimized. No prospective single-panel annotation campaign is planned within the v1.0 scope.

Reliance on the source’s continued availability. If the LIDC hosting institution withdraws the dataset, v0.5’s manifest still records what was included but the underlying scans become unavailable. The mirrored content hash (PhysioNet DOI and Zenodo) provides a permanent record of what v0.5 *was*; it does not guarantee the scans remain downloadable.

Data versioning and persistent identifiers

Following the FAIR Guiding Principles for scientific data management and stewardship (Wilkinson et al., 2016), each release of PWM-LDCT is uniquely identified by the SHA-256 hash of its `manifest.sha256` file, which itself hashes every released artifact (annotation files, simulated-low-dose images, harmonized metadata, splits, Dockerfiles, and schema). This content hash lets any downstream user verify offline that their local copy of the deposited artifacts is bit-identical to the release, and that shards they regenerate from their own source distributions match the release within the documented per-voxel tolerance, independent of any hosting institution remaining online.

The release is deposited and cited through standard, journal-recognized channels: the assigned **PhysioNet DOI** (10.13026/prefix-0.5.0) is the primary citation handle, and a versioned archive of the code and deposited records is mirrored to **Zenodo** (DOI to be assigned at submission) and the project’s GitHub release, with the same SHA-256 content hash recorded in each so the descriptor’s reproducibility claims do not depend on any single channel remaining available.

Releases follow MAJOR.MINOR.PATCH semantics: MAJOR breaks schema compatibility, MINOR adds sources or annotation tasks without breaking schema, and PATCH corrects errors (e.g., a scan flagged after PHI re-audit). The v1.0 validation subset (AAPM 2016) ships as a MINOR release with its own DOI and content hash.

Errata and data-correction policy. Errors (a missed PHI hit, a discordant annotation surviving QA, a reproducibility regression) are reported via GitHub issues (<https://github.com/integ>

[ritynoble/low_dose_CT/issues](#), label `errata-v0.5`) and batched into PATCH releases that increment the version, regenerate the content hash, and update the deposited record and its mirrors. Each PATCH ships an erratum note (what changed, which scans are affected) and a cumulative `ERRATA.md`. Prior versions remain resolvable by their content hashes, so papers citing an earlier release stay reproducible.

Planned v1.0 roadmap

v1.0 roadmap. The v1.0 roadmap adds the AAPM 2016 public source (10 training patients, Siemens, with full-dose and reference quarter-dose reconstructions plus projections) under the same harmonization schema, as a real-paired-dose *validation subset* rather than a scale expansion. The AAPM 2016 subset carries real paired-dose data (full-dose acquisitions; reduced-dose by Mayo’s validated noise insertion), top-up lung-nodule annotation on chest cases where available, and cross-source consistency / simulation-vs-reference validations against the LIDC simulated low-dose cohort. Mayo LDCT-PD and prospective clinical acquisition were removed from the release scope on 2026-08-21 and are not planned. The v1.0 release ships as a MINOR release with its own DOI and content hash.

Code Availability

- Dataset loader: https://pypi.org/project/pwm_ldct_loader/ (Apache 2.0)
- Preprocessing Dockerfile: https://github.com/integritynoble/low_dose_CT/tree/main/WS-1_dataset/pipelines/ (LIDC pipeline ships with v0.5; the AAPM pipeline is v1.0 validation subset)
- Annotation artifacts: https://github.com/integritynoble/low_dose_CT/tree/main/WS-1_dataset/annotations/
- Forward model used for low-dose simulation: https://github.com/integritynoble/low_dose_CT/tree/main/WS-1_dataset/pipelines/pwm_ldct_prep/lowdose_sim.py (reference implementation; defers to `pwm_core.contrib.modalities.ct_radon` when installed)
- Deposited records + manifest: the PhysioNet release record under DOI `10.13026/prefix-0.5.0`, archived at Zenodo (DOI to be assigned at submission). The manifest’s SHA-256 content hash is recorded in both.

Author contributions

To be completed at submission using the CRediT taxonomy: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing-Original Draft, Writing-Review & Editing, Visualization, Supervision, Project administration, Funding acquisition.

Competing interests

One or more authors are affiliated with the PWM Protocol Foundation, which contributed the open-source registry tooling used in preprocessing. All data records and code are openly licensed

(Table 3) and are fully obtainable, citable, and verifiable through the PhysioNet DOI and the Zenodo archive, so this affiliation confers no privileged access to the release. At submission, each author will declare any financial interest in the PWM Protocol Foundation (equity, consulting, or advisory role) or state its absence. The authors declare no other competing interests.

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Dimension	LIDC-IDRI (Armato III et al., 2011)	AAPM 2016 (AAPM, 2016)	Mayo LDCT-PD (McCollough et al., 2017)	NLST (National Lung Screening Trial Research Team, 2011; National Cancer Institute, 2013)	PWM-LDCT v0.5
Scanner-vendor coverage	4 vendors (GE, Siemens, Philips, Toshiba)	Siemens only	GE + Siemens ($\approx 50/50$)	Multi-site, multi-vendor (screening)	4 vendors (LIDC-derived)
Paired full/low dose	No (full-dose only)	Yes, 25% noise-inserted	Yes, 25% noise-inserted	No (single screening acquisition)	No real paired dose; simulated LD at 0.10/0.25/0.50 (v1.0 adds AAPM 10-patient real-paired validation)
Raw projections (sinograms)	No	Yes	Yes	No	No (v1.0 adds AAPM sinograms)
Sample size (patients)	1,010	10	199 (public)	26,722 (LDCT arm)	1,010 (v0.5); 1,020 planned for v1.0
Annotation depth	4-radiologist nodule labels	None as-distributed	Lesion labels on subset	No pixel-level labels (trial outcomes only)	Harmonized multi-task: nodule majority-vote + per-reader (roadmap: top-up + Likert in v1.0)
Dose protocol	Standard clinical dose	Projection-domain noise insertion at 25%	Projection-domain noise insertion at 25%	Fixed low-dose screening protocol (non-paired)	Physically-grounded simulated LD (Poisson-thinning); roadmap: real noise-inserted reference
Unified schema / content addressing	No	No	No	No	Yes (HDF5 shards + SHA-256 manifest)
Access model	Open (NBIA, CC BY 3.0)	Research DUA (Mayo)	Credentialed (TCIA)	Controlled (NCI CDAS)	PhysioNet DOI + Zenodo mirror (records CC BY 4.0)

Table 2: Comparison of PWM-LDCT v0.5 with existing public low-dose CT resources. “Paired full/low dose” refers to same-patient acquisitions or matched low-dose derivatives generated from the patient’s own full-dose projection data; NLST is a screening trial and does not provide paired-dose scans, pixel-level annotations, or raw projections. PWM-LDCT v0.5 harmonizes LIDC-IDRI (its sole source) into a content-addressed, multi-task benchmark; the AAPM 2016 row describes the resource that PWM-LDCT plans to harmonize as the v1.0 real-paired validation subset rather than a dataset included in this release, while the Mayo LDCT-PD row is retained for landscape context only (not in the release scope, removed 2026-08-21). The v1.0 plan of combining *cross-vendor* LIDC-derived simulated coverage with an AAPM 2016 (Siemens) *real-paired* validation subset, all under one harmonized schema, is unique to PWM-LDCT among public low-dose CT resources.

Path	Contents	License avail.	/
<i>(i) New data records — deposited in full</i>			
<code>annotations/lidc_majority_vote</code>	Majority-vote consolidation of the four-radiologist LIDC nodule labels	CC BY 3.0	
<code>annotations/raw_per_reader</code>	Per-reader raw labels (pre-majority-vote), for uncertainty studies	CC BY 4.0	
<code>sim_lowdose/lidc/</code>	Simulated low-dose images for the full-dose-only LIDC scans (HDF5), $r \in \{0.10, 0.25, 0.50\}$	CC BY 3.0	
<code>metadata/</code>	Harmonized per-scan <code>metadata.json</code> (acquisition params, HU offsets, kernel, flags)	CC BY 4.0	
<code>splits/{train,val,test}.txt</code>	LIDC patient IDs per split (seed 42)	CC BY 4.0	
<code>manifest.sha256</code>	SHA-256 of every released artifact (the content address)	CC0	
<i>(ii) Access & reproduction code</i>			
<code>pwm_ldct_loader</code> (PyPI)	Unified Python loader (LIDC in v0.5; AAPM in v1.0 validation subset)	Apache 2.0	
<code>pipelines/Dockerfile.lidc</code>	Deterministically regenerate harmonized HDF5 (within the released per-voxel tolerance) from the LIDC source's raw DICOM	Apache 2.0	
<code>schema/*.md</code>	Per-record schema, DICOM cleaning spec, DICOM→HDF5 mapping, annotation-QA protocol	Apache 2.0	
<i>(iii) Not redistributed — obtained from source, regenerated locally</i>			
Underlying DICOM scans	Full-dose LIDC images (NBIA)	LIDC CC BY 3.0 / NBIA terms	
AAPM scans + derivatives (v1.0 validation subset)	Full/low-dose images and sinograms; pixel derivatives of DUA-restricted scans	Not included in v0.5	

Table 3: PWM-LDCT v0.5 (LIDC-only) release contents, grouped by license and availability. The new data records (annotations, LIDC-derived simulated low-dose, harmonized metadata, splits, manifest) are deposited in full under open licenses; the access-and-reproduction code is Apache 2.0; the underlying DICOM scans are not redistributed — users obtain the LIDC scans from the NBIA gatekeeper and run the Dockerfile pipeline locally to regenerate HDF5 shards matching `manifest.sha256`. AAPM 2016, its real low-dose pairs, and its pixel derivatives are v1.0 validation-subset items and do not appear in v0.5; Mayo LDCT-PD is not in the release scope (removed 2026-08-21).

Source	Train	Val	Test	Total
LIDC-IDRI (v0.5)	589	216	205	1,010
<i>v1.0 validation subset</i>				
AAPM 2016	4	2	4	10

Table 4: v0.5 cohort composition by source and split (LIDC-only). Splits are patient-level and deterministic given the published seed (42); LIDC contributes 1,010 of its canonical 1,018 patients (8 are unavailable in the public distribution used here). The AAPM 2016 row is the v1.0 real-paired validation subset (not part of v0.5); Mayo LDCT-PD is not in the release scope (removed 2026-08-21).

Parameter	LIDC-IDRI (v0.5)	
Manufacturer set	GE, Siemens, Philips, Toshiba (670 / 201 / 73 / 66)	
Scanner model(s)	heterogeneous, per-scan in <code>metadata.json</code> (4-vendor)	Si
Tube voltage (kVp), typical	120 (813/1,010 of scans; tail 130–140)	
Full-dose effective mAs, typical	median 230 (range 0–11,200; IQR 32–943; site-heterogeneous)	
Low-dose ratio (paired)	N/A (full-dose only; v0.5 provides <i>simulated</i> low dose at 0.10/0.25/0.50)	
Pitch, typical	median 0.94 (recorded on 132 scans; site-heterogeneous)	
Slice thickness (mm), thin series	0.60–5.00 (median 2.0) across patients	
Reconstruction kernel(s)	heterogeneous (STANDARD, BONE, B30f, LUNG, and 14 further kernels), per scan	

Table 5: Acquisition parameters. v0.5 contains LIDC only ($n = 1,010$); the LIDC cells are regenerated from `metadata.json` by the committed extraction script `analysis/extract_acquisition.py` (build provenance: 1010 files, combined SHA-256 20476a8105319e6192eebf233e274a4f6f5f787a574bd605809963f3160df870). The AAPM 2016 column is the v1.0 real-paired validation subset and is not part of v0.5; Mayo LDCT-PD is not in the release scope (removed 2026-08-21). Full per-scan values — including the exact scanner-model string, pitch, kernel, and per-scan effective mAs — are exposed by the loader without imposing a single value. LIDC’s broad multi-vendor coverage (four manufacturers: GE, Siemens, Philips, Toshiba) is a noteworthy strength for cross-vendor generalization studies.

Field	LIDC-IDRI (v0.5)	AAPM 2016 (v1.0 validation subset)
N (patients, unique)	1,010	10
Age, median (IQR) [yr]	61 (50–70)*	N/A [†]
Age, range [yr]	14–88*	N/A [†]
Sex (M / F), n (%)	140 / 140 (28%)*	N/A [†]
BMI, median (IQR) [kg/m ²]	N/A	N/A
Scan-date range	N/A	N/A
Primary indication	Lung-cancer screening + nodule eval	Oncology (abdomen)
Geographic origin (sites)	US multi-site (LIDC consortium)	Mayo Clinic, Rochester MN

Table 6: Patient-level demographics. v0.5 contains LIDC only ($n = 1,010$); the AAPM 2016 column is the v1.0 real-paired validation subset and is not part of v0.5; Mayo LDCT-PD is not in the release scope (removed 2026-08-21). Age, sex, BMI, and scan-date fields are aggregated from the source’s published metadata where available; fields not preserved in a source’s de-identified release are recorded as N/A. The LIDC cells are regenerated from `metadata.json` by the committed extraction script `analysis/extract_demographics.py` (build provenance: 1010 files, combined SHA-256 20476a8105319e6192eebf233e274a4f6f5f787a574bd605809963f3160df870). *LIDC age and sex are recorded only on a subset of patients (age 20%, sex 28%) per LIDC’s variable DICOM-header preservation; the percentages in the Sex row are computed over the patients with known sex. [†]AAPM 2016 patients are fully anonymized (PatientID = “Anonymous”) in the public distribution; age and sex were stripped at de-identification and are not recoverable. BMI and StudyDate are not preserved in any of the source releases. Geographic origin captures the population from which each cohort is sampled and bears on generalizability claims.

Vendor	n_{series}	W_1 vs. aggregate (HU)	Spacing overlap (%)	Thickness overlap (%)
GE Medical Systems	30	114.5	48.3	29.0
Siemens	30	43.4	61.7	74.7
Toshiba	30	85.6	70.0	—
Philips	30	27.5	63.3	57.9

Table 7: Harmonization validation under v0.5 (LIDC-only): HU-histogram 1-Wasserstein distance to the aggregate (clinical window $[-1100, 500]$ HU), pixel-spacing distribution overlap, and slice-thickness distribution overlap, computed on a stratified random sample of 30 series per vendor (120 total, seed 42) from the project’s local harmonized LIDC root. The worst observed vendor pair ($W_1 = 199.3$ HU, GE–Toshiba) sets the 200 HU threshold quoted in the text. Toshiba contributes no thin (≤ 1.5 mm) series, hence the missing thickness overlap; the AAPM 2016 panel is a v1.0 validation-subset item, and Mayo LDCT-PD is not in the release scope. Regenerated by the temporary validation script `hv_fast.py` (intermediate provenance).

Case	Slice	Attribute	Published	Recomputed
lidc-0128	172	texture (1–5)	3	5
lidc-0128	173	texture (1–5)	3	5
lidc-0128	174	texture (1–5)	3	5
lidc-0128	175	texture (1–5)	3	5
lidc-0615	173	unrecorded cluster	–	6.62 mm (3 readers)
lidc-0007	112	diameter (mm)	21.484	17.188
lidc-0866	73	texture (1–5)	5	3
lidc-0606	197	diameter (mm)	18.246	13.516
lidc-0606	194	diameter (mm)	17.571	14.191
lidc-0622	142	diameter (mm)	21.723	18.480
lidc-0622	137	texture (1–5)	3	5
lidc-0699	234	diameter (mm)	17.619	7.957
lidc-0699	243	diameter (mm)	17.619	1.137
lidc-0699	231	unrecorded cluster	–	10.8 mm (3 readers)
lidc-0804	255	texture (1–5)	3	5
lidc-0804	254	texture (1–5)	3	5
lidc-0804	253	texture (1–5)	3	5

Table 8: Majority-vote reconciliation discrepancies on the 100-patient random sample. Unrecorded-cluster rows mark nodule-sized (≥ 3 mm) annotations agreed by ≥ 3 of 4 readers that are absent from the deposited majority-vote; diameter rows report the largest per-case deviation between the deposited value and the median of contributing readers; texture rows report label-mode differences > 1 step on the 1–5 scale. Reconciled by the temporary script `nodule_mv_reconcile.py` against `annotations/raw_per_reader/`.

Reader group	N cases	Detection κ (range)	Texture κ (range)	Pass ≥ 0.60
anon_1–anon_4 (primary)	677	0.926–0.997	0.330–0.471	detection: all pairs; texture: no
reader1–reader4	21	0.243–0.751	0.164–0.817	mixed
anonymous (numeric IDs)	6	0.748–1.000	0.000–0.444	detection: all pairs*

Table 9: **Inter-rater annotation reliability.** Pairwise reader agreement computed by `pipelines/pwm_ldct_prep/interrater_metrics.py` over the LIDC `raw_per_reader` files (detection Cohen’s κ after IoU ≥ 0.30 matching, per §3.1 of `schema/annotation_qa_protocol.md`; texture κ = linearly-weighted Cohen’s κ on the 1–5 ordinal scale over matched nodules). Reader groups correspond to four non-homogeneous reader-ID naming schemes present in the raw release (*anon*, *reader*, *anonymous*, numeric); groups are therefore reported separately and are not merged. Likert ICC(2,k) is not reported because no Likert series exist in the current release. *The anonymous group’s texture κ includes 0.000 values for three reader pairs, consistent with a reader-ID naming artifact in this small group ($n = 6$); detection agreement in the same group is high.

Model	r010			r025			r050		
	PSNR (dB)	SSIM	LPIPS	PSNR (dB)	SSIM	LPIPS	PSNR (dB)	SSIM	LPIPS
RED-CNN	51.96	0.9893	0.0053	53.65	0.9929	0.0026	54.46	0.9941	0.0020
LEARN	53.99	0.9912	0.0053	57.06	0.9958	0.0024	58.96	0.9974	0.0018
CTformer	51.12	0.9882	0.0073	53.80	0.9930	0.0037	55.33	0.9946	0.0027
CoreDiff	51.59	0.9877	0.0057	53.36	0.9921	0.0025	54.25	0.9934	0.0017

Table 10: **Baseline reproductions.** Test-split PSNR / SSIM / LPIPS for four baselines on PWM-LDCT v0.5 (LIDC-only), per simulated dose level (0.10/0.25/0.50), computed with the released Docker containers. PSNR and SSIM follow standard definitions (Wang et al., 2004) and LPIPS is a learned perceptual metric (Zhang et al., 2018). All four rows were evaluated on the same $n = 764$ test slices (the CTformer row uses the retrained compact checkpoint, Table 11). Following the field rule that a fidelity number is reported only beside a task-based detectability number (§4 of the companion charter), Table 11 reports the paired detectability assessment on the same task.

Model	r010		r025		r050		task & observer		
	PSNR	CNR	PSNR	CNR	PSNR	CNR	task	location	ob
RED-CNN	51.96	5.19	53.65	8.05	54.46	10.97	SKE-Gaussian20HU-s2px	known	CHO(D
LEARN	53.99	6.50 [†]	57.06	12.24 [†]	58.96	22.21 [†]	SKE-Gaussian20HU-s2px	known	CHO(D
CTformer	52.93	2.57	55.32	4.61	56.63	6.86	SKE-Gaussian20HU-s2px	known	CHO(D
CoreDiff	51.52	5.34	53.36	8.43	54.28	11.54	SKE-Gaussian20HU-s2px	known	CHO(D
Gaussian blur (trap)	41.62	5.75	41.65	9.38	41.65	13.04	SKE-Gaussian20HU-s2px	known	CHO(D

Table 11: **Paired fidelity + task-based detectability.** The detectability task is signal-known-exactly (SKE): a 2D Gaussian lesion ($\sigma = 2$ px, peak contrast 20 HU) inserted into the low-dose input at a known location, reconstructed by each method, and scored on a tissue-only ROI (HU band 10–120, low gradient) that contains no anatomy. CNR is computed against the Rose criterion of 3.0 (reliable detection) (Rose, 1948; Burgess, 1999); CHO AUC was also computed with four difference-of-Gaussians channels and no internal noise, and saturated at 1.000 for every method on this SKE task, so CNR is the discriminating index. The **Gaussian blur** (fixed $\sigma = 1.0$, 5×5 kernel) is a permanent control: it scores the lowest PSNR (41.6–41.7 dB) of all methods while its CNR (5.75–13.04) exceeds every learned baseline with a computable mean at every dose level and crosses the Rose criterion even at r010, demonstrating that fidelity (PSNR) and detectability (CNR) are decoupled and that a high CNR produced by smoothing alone is not diagnostic content (the blur trap; BandER separates it by 8.9–15.6 $\times$ at the ROI level on the AAPM held-out 4 patients (192 ROI/model); Table 14). Detectability numbers here use the full $n = 764$ test slices per dose level (aggregate mean over the fixed 5-seed set, the same $n = 764$ as Table 10); [†]LEARN’s slice-wise CNR mean is undefined because some output slices have zero local noise variance (division by zero, identical across the 5-seed grid), so its CNR is reported as the slice median for transparency.

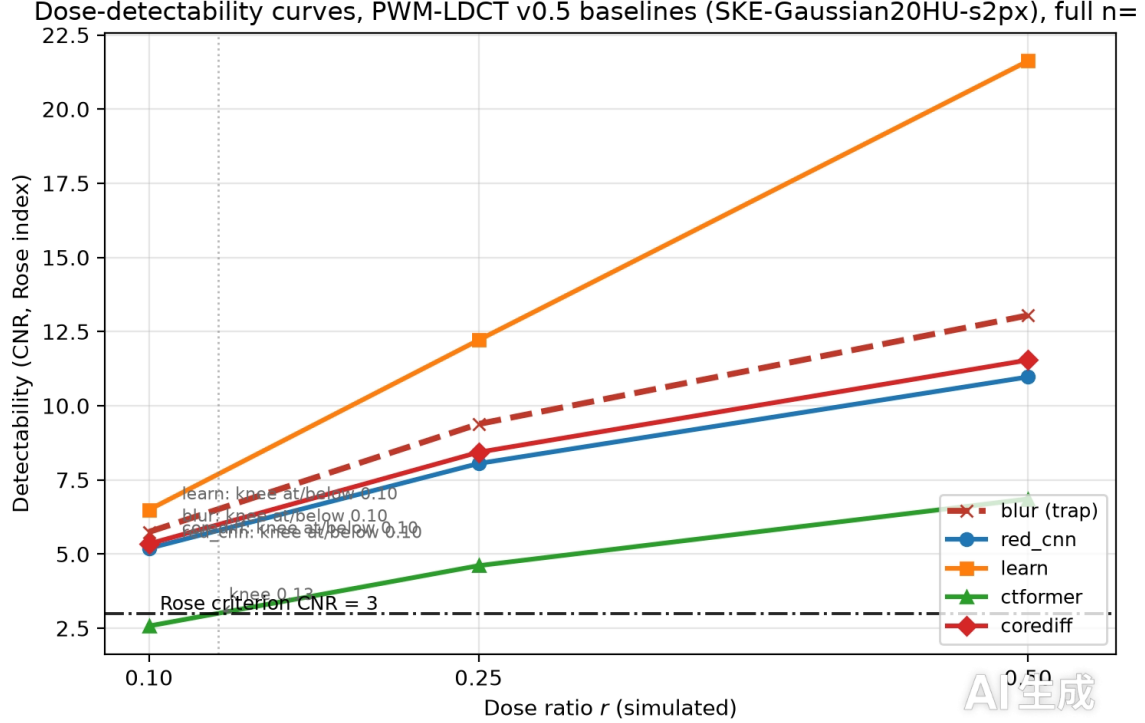


Figure 4: **Dose-detectability curves.** CNR versus simulated dose ratio for the four baselines and the standing Gaussian-blur trap, with the Rose criterion ($\text{CNR} = 3.0$) drawn as the reference line; each point is the 5-seed aggregate over the full $n = 764$ test slices per dose level (LEARN is reported as the slice median because its slice-wise mean is undefined, Table 11). Under this full-scale evaluation every method except CTformer already exceeds the Rose criterion at the lowest simulated dose (RED-CNN CNR 5.19, LEARN 6.50, CoreDiff 5.34, blur 5.75 at r_{010}), so their Rose-crossing knee lies at or below $r = 0.10$, the lowest sampled point; CTformer is the only method whose knee falls inside the measured range, crossing the Rose criterion between r_{010} and r_{025} at dose ratio ≈ 0.13 ($\text{CNR } 2.57 \rightarrow 4.61$). The full-scale curve thus shows that a CNR-only criterion cannot order the learned baselines against the smoothing-only blur trap — all four already cross Rose at the lowest simulated dose — and re-affirms the need for the frequency-domain dimension (BandER) as the discriminating index. Single-point detectability claims are replaced by a curve statement: the dose at which each method becomes reliably detectable is exactly the quantity the dataset is designed to expose (companion charter, Rung 4). The curves were produced by `baselines/src/pwm_ldct_baselines/dose_curve.py` from the committed full- n per-model result JSONs and the corresponding statistics are released in `output/dose_detectability_stats.json`.

Model	Observed knee	95% CI	Not reached ($B = 4,000$)	Remark
RED-CNN	≤ 0.10	≤ 0.10	0	r010 CNR 5.19 > Rose 3.0; all 4,000 d
LEARN	≤ 0.10	≤ 0.10	0	slice median; r010 CNR 6.50
CoreDiff	≤ 0.10	≤ 0.10	0	r010 CNR 5.34 > Rose
CTformer	0.131	[0.123, 0.139]	0	crosses Rose between r010
Gaussian blur (trap)	≤ 0.10	≤ 0.10	0	full per-slice data

Table 12: **Slice-level bootstrap confidence intervals for the simulated dose-detectability knee (LIDC v0.5).** Knee = dose ratio at which the interpolated CNR crosses the Rose criterion (3.0) on the fixed grid $\{0.10, 0.25, 0.50\}$; slice-level resampling (with replacement, the 764 distinct test slices per dose level), $B = 4,000$ percentile 95% CIs, fixed RNG seed (20260831), regenerated by `analysis/bootstrap_lidc_sim.py`. LEARN’s CNR is the slice median (its slice-wise mean is undefined; Table 11). A knee reported as ≤ 0.10 means the r010 point already exceeds the Rose criterion in every draw, so the knee lies at or below the lowest simulated dose; no extrapolation is attempted outside the measured range. Resampling details and per-dose CNR CIs in `output/lidc_simulated_bootstrap_report.md`.

Model	Fidelity			Detectability (BandER)		Insertion (transparent)	
	PSNR (dB)	SSIM	LPIPS	ROI	full	CNR	CHO AUC
RED-CNN	41.2	0.934	—	3.876	1.524	0.10	1.000
CTformer	39.6	0.912	—	6.735	2.197	0.09	1.000
LEARN	41.6	0.936	—	3.854	1.514	0.10	1.000
Gaussian blur (trap)	41.6	0.965	—	0.432	0.430	0.17	1.000

Table 13: **Real-paired noise-ROI detectability on the AAPM 2016 held-out test (R3).** Per-patient means over 4 patients (48 ROI/patient; 192 ROI/model on the test volume). ROI BandER is the frequency-domain retention ratio in the 1-px high-pass band of real anatomy inside the tissue ROI; blur separates from every learned baseline by 9–15 $\times$ while holding the highest SSIM (0.965) and top PSNR (41.6 dB) — the high-fidelity/low-detectability trap on real tissue. Insertion-based indices (CNR, CHO AUC, NPWE) remain non-discriminative (CHO AUC saturated at 1.000 for all methods), reported transparently per the R1 calibration conclusion.

Index	blur value	model range (min–max)	separation	blur worst?	Identifies t
ROI BandER (freq.)	0.432	3.854–6.735	8.9–15.6 $\times$	Yes (always)	Yes
CNR	0.17	0.09–0.10	1.6–1.9 $\times$ (reversed)	No (blur highest)	No
CHO (AUC)	1.000	1.000	1.0 $\times$ (saturated)	No (no separation)	No
NPWE	220,805	222,630–223,915	1.008 $\times$ (< 1%)	Borderline (< 1%)	No
TM-AUC (reference)	0.990	0.990	1.0 $\times$	No	No

Table 14: **Discrimination power of candidate detectability indices on real anatomy (R3).** Aggregated per-patient means over the 4 held-out patients (192 ROI/model on the test volume). Separation = model range / blur value (BandER and NPWE) or absolute contrast (CNR, AUC); “reversed” means the blur trap scores *higher* than the models, i.e. the index fails to flag it. ROI BandER is the only index that separates the trap by $\geq 8.9\times$, ranks it worst in all 4,000 slice-level bootstrap draws ($P(\text{blur last}) = 1.0000$), and passes the paired binomial sign test ($p \approx 1.6 \times 10^{-58}$, 192/192 aligned slices). Statistics from `output/aapm_observer_sensitivity_real.json`.

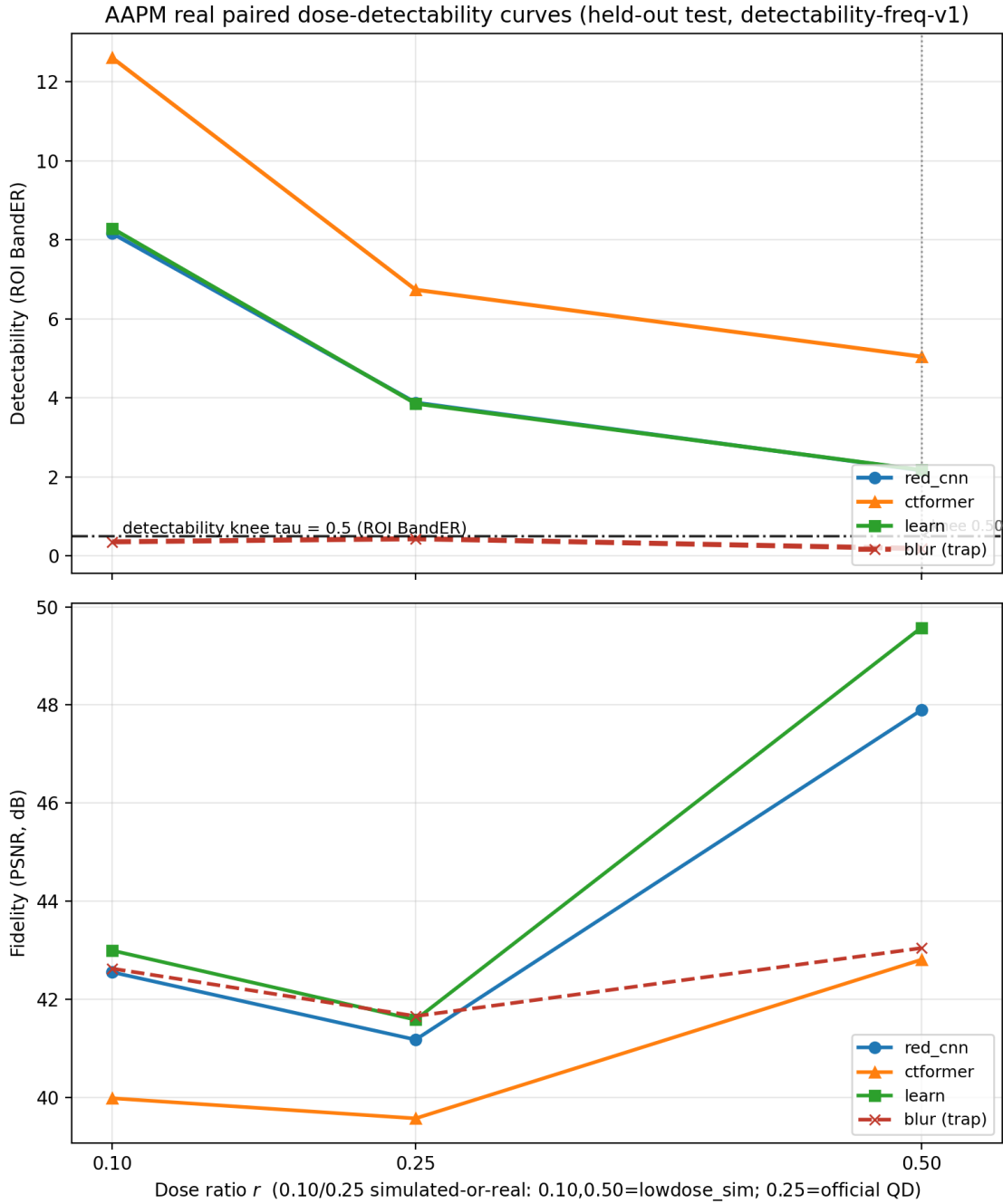


Figure 5: **Real-paired dose-detectability curve (AAPM 2016 held-out test, R4).** ROI BandER (detectability-freq-v1, mean over 48 slices/patient $\times$ 4 patients) versus dose ratio for the three learned baselines and the permanent Gaussian-blur trap. Dose points: $r = 0.10/0.50$ simulated with the release forward model; $r = 0.25$ official real QD. Knee ($\tau = 0.5$, dashed) is reached at $r = 0.500$ by RED-CNN / CTformer / LEARN and not reached by blur; compare the LIDC simulation-based CNR knee (Figure 4: at/below 0.10 for RED-CNN/LEARN/CoreDiff/blur, ≈ 0.13 for CTformer) — the real-data knee is right-shifted. Produced by the analysis pipeline; released as `output/aapm_dose_detectability_curve.png` + `aapm_dose_detectability.json`.

Group	blur ROI BandER	min model ROI BandER	ratio	blur rank
<i>Per vendor (LIDC simulated $r = 0.25$; AAPM real QD)</i>				
GE (sim)	0.045	0.438 (LEARN)	$9.7\times$	4/4
Philips (sim)	0.249	3.193 (RED-CNN)	$12.8\times$	4/4
Siemens (sim)	0.122	1.157 (RED-CNN)	$9.5\times$	4/4
Toshiba (sim)	0.059	1.123 (LEARN)	$19.0\times$	4/4
AAPM-Siemens (real QD)	0.432	3.854 (LEARN)	$8.9\times$	4/4
<i>Per dose (AAPM-Siemens; sim = forward model, real = official QD)</i>				
$r = 0.10$ (sim)	0.353	8.164 (RED-CNN)	$23.1\times$	4/4
$r = 0.25$ (real QD)	0.432	3.854 (LEARN)	$8.9\times$	4/4
$r = 0.50$ (sim)	0.183	2.163 (RED-CNN)	$11.8\times$	4/4

Table 15: **Blur-trap separation across vendor and dose groups (R5/R6).** Group-mean ROI BandER for the blur trap versus the weakest learned baseline within each group (ratio = min-model/blur), and blur’s detectability rank within the group. All 8 groups pass the $\geq 3\times$ separation gate; blur ranks last (4/4) everywhere. GE/Toshiba groups contain near-zero full-dose high-frequency denominators in selected ROIs; an adaptive epsilon floor ($10^{-4}\times$ full-image FD energy, 2026-08-22) removes the resulting order-of-magnitude inflation, so ROI BandER is comparable across vendors at the 10^0 scale while rank-based statistics and separation ratios are unchanged. Group statistics: vendor effect permutation KW $p = 0.0103$ (all 4 models); dose effect exact-permutation Friedman $p = 0.0417$ (3/4 models; CTformer $p = 0.1250$). Released as `output/aapm_lidc_cross_vendor_spread.json` and `output/aapm_per_dose_spread.json`.

Model	Fidelity channel		Detectability channel	
	output SSIM	Δr_{SSIM}	output BandER	$\varepsilon_{\text{BandER}}$
RED-CNN	0.934	0.258	3.876	0.096
CTformer	0.912	0.531	6.735	0.568
LEARN	0.936	0.263	3.854	0.101
Gaussian blur (trap)	0.965	0.693	0.432	0.898

Table 16: **Predicted vs. measured equivalent dose on AAPM real paired data (WS-2).** Per-patient means over the 4 held-out patients ($r^* = 0.250$ verified against the official QD reference). The detectability contrast channel $\varepsilon_{\text{BandER}}$ identifies the blur trap at 0.898 versus 0.096–0.568 for learned baselines, while the fidelity channel shows blur with the highest output SSIM (0.965) and only a moderate $\Delta r_{\text{SSIM}} = 0.693$ — the fidelity-masking phenomenon. Physical input BandER is non-monotone in dose and is reported only as a contrast channel; full per-patient detail in `output/aapm_pred_vs_measured_equiv_dose_summary.md`.